

MTH251 & MTH252

# **Mathematical Analysis I & II**

Course Textbook

Department of Mathematical Sciences  
**UNIST**

## Mathematical Analysis I & II

Course Textbook

### Author

Muho Choi (Department of Computer Science and Engineering)

Email: [sihyeon8240@unist.ac.kr](mailto:sihyeon8240@unist.ac.kr)

GitHub: <https://github.com/sihyeon8240>

### Source Code & Contributions

GitHub: <https://github.com/sihyeon8240/Math>

Feel free to open issues or pull requests for corrections.

© 2026. Muho Choi. All rights reserved.

Developed for the Department of Mathematical Sciences, UNIST.

This textbook is licensed under a **Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License** (CC BY-NC-SA 4.0).

**You are free to:** Share, copy, and redistribute the material in any medium or format; adapt, remix, transform, and build upon the material.

**Under the following terms:** You must give appropriate credit; you may not use the material for commercial purposes; if you remix or transform the material, you must distribute your contributions under the same license.

# Contents

|          |                                                             |           |
|----------|-------------------------------------------------------------|-----------|
| <b>1</b> | <b>The Number Systems</b>                                   | <b>1</b>  |
| 1.1      | The Natural, Integer, and Rational Number Systems . . . . . | 1         |
| 1.2      | Ordered Sets and Ordered Fields . . . . .                   | 2         |
| 1.3      | The Real Field . . . . .                                    | 5         |
| 1.4      | The Extended Real Number System . . . . .                   | 13        |
| 1.5      | The Complex Field . . . . .                                 | 14        |
| 1.6      | Euclidean Spaces . . . . .                                  | 17        |
| <b>2</b> | <b>Basic Topology</b>                                       | <b>19</b> |
| 2.1      | Relations . . . . .                                         | 19        |
| 2.2      | Functions . . . . .                                         | 20        |
| 2.3      | Finite Sets . . . . .                                       | 23        |
| 2.4      | Countable and Uncountable Sets . . . . .                    | 31        |
| 2.5      | Metric Spaces . . . . .                                     | 39        |
| 2.6      | Compact Sets . . . . .                                      | 39        |
| 2.7      | Perfect Sets . . . . .                                      | 39        |
| 2.8      | Connected Sets . . . . .                                    | 39        |
| <b>3</b> | <b>Numerical Sequences and Series</b>                       | <b>40</b> |
| 3.1      | Convergent Sequences . . . . .                              | 40        |
| 3.2      | Subsequences . . . . .                                      | 40        |
| 3.3      | Cauchy Sequences . . . . .                                  | 40        |
| 3.4      | Upper and Lower Limits . . . . .                            | 40        |
| 3.5      | Some Special Sequences . . . . .                            | 40        |
| 3.6      | Series . . . . .                                            | 40        |
| 3.7      | Series of Nonnegative Terms . . . . .                       | 40        |
| 3.8      | The Number $e$ . . . . .                                    | 40        |
| 3.9      | The Root and Ratio Tests . . . . .                          | 40        |
| 3.10     | Power Series . . . . .                                      | 40        |
| 3.11     | Summation by Parts . . . . .                                | 41        |
| 3.12     | Absolute Convergence . . . . .                              | 41        |
| 3.13     | Addition and Multiplication of Series . . . . .             | 41        |
| 3.14     | Rearrangements . . . . .                                    | 41        |

|          |                                                           |           |
|----------|-----------------------------------------------------------|-----------|
| <b>4</b> | <b>Continuity</b>                                         | <b>42</b> |
| 4.1      | Limits of Functions . . . . .                             | 42        |
| 4.2      | Continuous Functions . . . . .                            | 42        |
| 4.3      | Continuity and Compactness . . . . .                      | 42        |
| 4.4      | Continuity and Connectedness . . . . .                    | 42        |
| 4.5      | Discontinuities . . . . .                                 | 42        |
| 4.6      | Monotonic Functions . . . . .                             | 42        |
| 4.7      | Infinite Limits and Limits at Infinity . . . . .          | 42        |
| <b>5</b> | <b>Differentiation</b>                                    | <b>43</b> |
| 5.1      | The Derivative of a Real Function . . . . .               | 43        |
| 5.2      | Mean Value Theorems . . . . .                             | 43        |
| 5.3      | The Continuity of Derivatives . . . . .                   | 43        |
| 5.4      | L'Hospital's Rule . . . . .                               | 43        |
| 5.5      | Derivatives of Higher Order . . . . .                     | 43        |
| 5.6      | Taylor's Theorem . . . . .                                | 43        |
| 5.7      | Differentiation of Vector-valued Functions . . . . .      | 43        |
| <b>6</b> | <b>The Riemann-Stieltjes Integral</b>                     | <b>44</b> |
| 6.1      | Definition and Existence of the Integral . . . . .        | 44        |
| 6.2      | Properties of the Integral . . . . .                      | 44        |
| 6.3      | Integration and Differentiation . . . . .                 | 44        |
| 6.4      | Integration of Vector-valued Functions . . . . .          | 44        |
| 6.5      | Rectifiable Curves . . . . .                              | 44        |
| <b>7</b> | <b>Sequences and Series of Functions</b>                  | <b>45</b> |
| 7.1      | Discussion of Main Problem . . . . .                      | 45        |
| 7.2      | Uniform Convergence . . . . .                             | 45        |
| 7.3      | Uniform Convergence and Continuity . . . . .              | 45        |
| 7.4      | Uniform Convergence and Integration . . . . .             | 45        |
| 7.5      | Uniform Convergence and Differentiation . . . . .         | 45        |
| 7.6      | Equicontinuous Families of Functions . . . . .            | 45        |
| 7.7      | Equicontinuous Families of Functions . . . . .            | 45        |
| 7.8      | The Stone-Weierstrass Theorem . . . . .                   | 45        |
| <b>8</b> | <b>Some Special Functions</b>                             | <b>46</b> |
| 8.1      | Power Series . . . . .                                    | 46        |
| 8.2      | The Exponential and Logarithmic Functions . . . . .       | 46        |
| 8.3      | The Trigonometric Functions . . . . .                     | 46        |
| 8.4      | The Algebraic Completeness of the Complex Field . . . . . | 46        |
| 8.5      | Fourier Series . . . . .                                  | 46        |
| 8.6      | The Gamma Function . . . . .                              | 46        |

|           |                                                |           |
|-----------|------------------------------------------------|-----------|
| <b>9</b>  | <b>Functions of Several Variables</b>          | <b>47</b> |
| 9.1       | Linear Transformations . . . . .               | 47        |
| 9.2       | Differentiation . . . . .                      | 47        |
| 9.3       | The Contraction Principle . . . . .            | 47        |
| 9.4       | The Inverse Function Theorem . . . . .         | 47        |
| 9.5       | The Implicit Function Theorem . . . . .        | 47        |
| 9.6       | The Rank Theorem . . . . .                     | 47        |
| 9.7       | Determinants . . . . .                         | 47        |
| 9.8       | Derivatives of Higher Order . . . . .          | 47        |
| 9.9       | Differentiation of Integrals . . . . .         | 47        |
| <b>10</b> | <b>Integration of Differential Forms</b>       | <b>48</b> |
| 10.1      | Integration . . . . .                          | 48        |
| 10.2      | Primitive Mappings . . . . .                   | 48        |
| 10.3      | Partitions of Unity . . . . .                  | 48        |
| 10.4      | Change of Variables . . . . .                  | 48        |
| 10.5      | Differential Forms . . . . .                   | 48        |
| 10.6      | Simplexes and Chains . . . . .                 | 48        |
| 10.7      | Stokes' Theorem . . . . .                      | 48        |
| 10.8      | Closed Forms and Exact Forms . . . . .         | 48        |
| 10.9      | Vector Analysis . . . . .                      | 48        |
| <b>11</b> | <b>The Lebesgue Theory</b>                     | <b>49</b> |
| 11.1      | Set Functions . . . . .                        | 49        |
| 11.2      | Construction of the Lebesgue Measure . . . . . | 49        |
| 11.3      | Measure Spaces . . . . .                       | 49        |
| 11.4      | Measurable Functions . . . . .                 | 49        |
| 11.5      | Simple Functions . . . . .                     | 49        |
| 11.6      | Integration . . . . .                          | 49        |
| 11.7      | Comparison with the Riemann Integral . . . . . | 49        |
| 11.8      | Integration of Complex Functions . . . . .     | 49        |
| 11.9      | Functions of Class $\mathcal{L}^2$ . . . . .   | 49        |

# Chapter 1

## The Number Systems

### 1.1 The Natural, Integer, and Rational Number Systems

We denote the set  $\{1, 2, 3, \dots\}$  of natural numbers by  $\mathbb{N}$ . The set  $\mathbb{N}$  is defined by the following axioms:

- $1 \in \mathbb{N}$ .
- If  $n \in \mathbb{N}$ , then  $s(n) \in \mathbb{N}$  where  $s(n) = n + 1$  is called the **successor** of  $n$ .
- 1 is not the successor of any element of  $\mathbb{N}$ .
- If  $m, n \in \mathbb{N}$  and  $s(m) = s(n)$ , then  $m = n$ .
- Any subset of  $\mathbb{N}$  which contains 1, and contains  $s(n)$  whenever it contains  $n$ , must be equal to  $\mathbb{N}$ .

The axioms above are called the **Peano axioms** (or *Peano Postulates*).

*Remark.* Let  $P(n)$  be a statement about  $n \in \mathbb{N}$  and  $S := \{n \in \mathbb{N} \mid P(n) \text{ is true}\}$ . Suppose  $P(n)$  satisfies the following conditions:

- $P(1)$  is true.
- $P(n + 1)$  is true whenever  $P(n)$  is true.

This implies that

- $1 \in S$ .
- $n + 1 \in S$  whenever  $n \in S$ .

Thus  $S = \mathbb{N}$ , i.e.,  $P(n)$  is true for all  $n \in \mathbb{N}$ . This method of proof is called the **principle of mathematical induction**.

**Theorem 1.1.1.** Every nonempty subset of  $\mathbb{N}$  has a least element, i.e., if  $E \subseteq \mathbb{N}$  and  $E \neq \emptyset$ , then there exists an element  $m \in E$  such that

$$m \leq n \text{ for all } n \in E.$$

This property is called the **well-ordering property** of  $\mathbb{N}$ .

*Proof.* Let  $S$  be a nonempty subset of  $\mathbb{N}$  which does not have a least element. We define a set  $T := \mathbb{N} \setminus S$ , and  $L := \{n \in \mathbb{N} \mid \{1, 2, \dots, n\} \subseteq T\}$ . We will show that  $L = \mathbb{N}$ , which implies that  $S = \emptyset$ , a contradiction, by mathematical induction.

- **Base case:**  $1 \in L$  since  $\{1\} \subseteq T$ , otherwise  $1 \in S$  and 1 is the least element of  $S$ .
- **Inductive step:** Suppose  $n \in L$ , i.e.,  $\{1, 2, \dots, n\} \subseteq T$ . Then  $n + 1 \notin S$ , otherwise  $n + 1$  is the least element of  $S$ . Therefore,  $n + 1 \in T$ , and  $\{1, 2, \dots, n + 1\} \subseteq T$ , i.e.,  $n + 1 \in L$ .

By the principle of mathematical induction,  $L = \mathbb{N}$ . □

*Remark.* Now, we can define the set  $\mathbb{Z}$  of integers and the set  $\mathbb{Q}$  of rational numbers as follows:

- $\mathbb{Z} := \{m - n \mid m, n \in \mathbb{N}\}.$
- $\mathbb{Q} := \left\{ \frac{m}{n} \mid m \in \mathbb{Z}, n \in \mathbb{N} \right\}.$

Also, we denote the set of positive numbers or negative numbers as follows:

- $\mathbb{Z}^+ := \{n \mid n \in \mathbb{N}\}, \mathbb{Z}^- := \{-n \mid n \in \mathbb{N}\}.$
- $\mathbb{Q}^+ := \left\{ \frac{m}{n} \mid m \in \mathbb{N}, n \in \mathbb{N} \right\}, \mathbb{Q}^- := \left\{ \frac{m}{n} \mid m \in \mathbb{Z}^-, n \in \mathbb{N} \right\}.$

## 1.2 Ordered Sets and Ordered Fields

An **ordered set** is a set  $S$  together with a relation  $<$  such that for any  $x, y, z \in S$ ,

(A)  $x < y$  and  $y < z$  implies  $x < z$ .

(B) Only one of the relations  $x < y$ ,  $x = y$ , or  $x > y$  holds.

$x > y$  means  $y < x$ ,  $x \leq y$  means  $(x < y \text{ or } x = y)$  and  $x \geq y$  means  $y \leq x$ .

*Remark.* In an ordered set  $S$ , the relation  $<$  is called a **order** (or a **total order**) on  $S$  and the set  $S$  is called **totally ordered**. If a relation  $<$  satisfies (A) only, then the relation  $<$  is called a **partial order** on  $S$  and the set  $S$  is called **partially ordered**.

A set  $F$  is called a **field** if it is equipped with two operations, *addition* and *multiplication*, and satisfies the following axioms:

(A) Axioms for addition

- $x + y \in F$  for all  $x, y \in F$ .
- $x + y = y + x$  for all  $x, y \in F$ .
- $(x + y) + z = x + (y + z)$  for all  $x, y, z \in F$ .
- There exists an element  $0 \in F$  such that  $x + 0 = x$  for all  $x \in F$ .
- For each  $x \in F$ , there exists an element  $-x \in F$  such that  $x + (-x) = 0$ .

(B) Axioms for multiplication

- $x \cdot y \in F$  for all  $x, y \in F$ .
- $x \cdot y = y \cdot x$  for all  $x, y \in F$ .
- $(x \cdot y) \cdot z = x \cdot (y \cdot z)$  for all  $x, y, z \in F$ .
- There exists an element  $1 \in F$  with  $1 \neq 0$  such that  $x \cdot 1 = x$  for all  $x \in F$ .
- For each  $x \in F$  with  $x \neq 0$ , there exists an element  $x^{-1} \in F$  such that  $x \cdot x^{-1} = 1$ .

(C) The distributive law

- $x \cdot (y + z) = x \cdot y + x \cdot z$  for all  $x, y, z \in F$ .

A set  $F$  is called an **ordered field** if it is both an ordered set and a field, and satisfies the following axioms:

- If  $x, y, z \in F$  and  $x < y$ , then  $x + z < y + z$ .
- If  $x, y \in F$  and  $x > 0$  and  $y > 0$ , then  $x \cdot y > 0$ .

**Definition 1.2.1.** Let  $S$  be an ordered set and  $E \subseteq S$ .

- $E$  is **bounded above** if there exists  $M \in S$  such that  $x \leq M$  for all  $x \in E$ . Such an  $M$  is called an **upper bound** of  $E$ .
- $E$  is **bounded below** if there exists  $m \in S$  such that  $m \leq x$  for all  $x \in E$ . Such an  $m$  is called a **lower bound** of  $E$ .

**Theorem 1.2.2.** Any nonempty subset of  $\mathbb{Z}$  which is bounded below has a unique least element, and any nonempty subset of  $\mathbb{Z}$  which is bounded above has a unique greatest element.

*Proof.* Let  $E \subseteq \mathbb{Z}$  be nonempty and bounded below. Then there exists  $m \in \mathbb{Z}$  such that  $m \leq x$  for all  $x \in E$ . Let  $F := \{x - m + 1 \mid x \in E\} \subseteq \mathbb{N}$ . By [Theorem 1.1.1](#),  $F$  has a least element  $n$ .

$$n \leq x - m + 1 \text{ for all } x \in E \implies n + m - 1 \leq x \text{ for all } x \in E.$$

Since  $n \in F$ , there exists a unique  $x_0 \in E$  such that  $n = x_0 - m + 1$ , so  $x_0 = n + m - 1$  is the unique least element of  $E$ .

Let  $S \subseteq \mathbb{Z}$  be nonempty and bounded above. Then there exists  $M \in \mathbb{Z}$  such that  $x \leq M$  for all  $x \in S$ . Let  $T := \{-x + M + 1 \mid x \in S\} \subseteq \mathbb{N}$ . By [Theorem 1.1.1](#),  $T$  has a least element  $n$ .

$$n \leq -x + M + 1 \text{ for all } x \in S \implies M - n + 1 \geq x \text{ for all } x \in S.$$

Since  $n \in T$ , there exists a unique  $x_0 \in S$  such that  $n = -x_0 + M + 1$ , so  $x_0 = M - n + 1$  is the unique greatest element of  $S$ .  $\square$

**Definition 1.2.3.** Let  $S$  be an ordered set and  $E \subseteq S$ . Suppose  $E$  is bounded above. Suppose there exists an  $\alpha \in S$  such that

- $\alpha$  is an upper bound of  $E$ .
- If  $\beta < \alpha$ , then  $\beta$  is not an upper bound of  $E$ .

Then  $\alpha$  is called the *least upper bound* or **supremum** of  $E$ , denoted by

$$\alpha = \sup E.$$

Similarly, suppose  $E$  is bounded below. Suppose there exists a  $\alpha \in S$  such that

- $\alpha$  is a lower bound of  $E$ .
- If  $\beta > \alpha$ , then  $\beta$  is not a lower bound of  $E$ .

Then  $\alpha$  is called the *greatest lower bound* or **infimum** of  $E$ , denoted by

$$\alpha = \inf E.$$

**Example 1.2.4.** Let  $S := \{p \in \mathbb{Q}^+ \mid p^2 < 2\}$ . For any  $p \in S$ , let  $q = p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2} > p$ . Then  $q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2} < 0$ , so  $q \in S$ . This implies that any  $p \in S$  is not an upper bound of  $S$ .

For any  $p \in \mathbb{Q}^+$  with  $p^2 > 2$ ,  $p$  is an upper bound of  $S$ . Let  $q = p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2} < p$ . Then  $q \in \mathbb{Q}^+$  and  $q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2} > 0$ , so  $q$  is an upper bound of  $S$ . This implies that such  $p$  is not the least upper bound of  $S$ .

Since there exists no  $p \in \mathbb{Q}$  such that  $p^2 = 2$ , the set  $S$  has no least upper bound in  $\mathbb{Q}^+$ .

**Definition 1.2.5.** An ordered set  $S$  is said to have the **least upper bound property** if every nonempty subset  $E \subseteq S$  that is bounded above has a least upper bound in  $S$ .

**Theorem 1.2.6.** If  $S$  is an ordered set with the least upper bound property, then any nonempty subset  $E \subseteq S$  that is bounded below has a greatest lower bound in  $S$ .

*Proof.* Let  $E$  be a nonempty subset of  $S$  that is bounded below. Let  $L$  be the set of all lower bounds of  $E$ . Then  $L$  is nonempty and bounded above by any element of  $E$ . Let  $\alpha = \sup L$ . We claim that  $\alpha = \inf E$ .

Suppose  $\alpha \notin L$ . Then there exists  $\beta \in E$  such that  $\beta < \alpha$ . Since  $\alpha = \sup L$ , there exists  $\gamma \in L$  such that  $\beta < \gamma \leq \alpha$ . But  $\gamma$  is a lower bound of  $E$ , so  $\gamma \leq \beta$ , a contradiction. Therefore,  $\alpha \in L$ .

Since  $\alpha = \sup L$ , for any  $\delta > \alpha$ ,  $\delta \notin L$ . Therefore,  $\alpha = \inf E$ . □

### 1.3 The Real Field

There exists an ordered field  $\mathbb{R}$  which has the **least upper bound property**. Moreover,  $\mathbb{R}$  contains  $\mathbb{Q}$  as a subfield, i.e.,  $\mathbb{Q} \subseteq \mathbb{R}$  and the operations of addition and multiplication on  $\mathbb{R}$  coincide with those on  $\mathbb{Q}$ .

We can construct  $\mathbb{R}$  from  $\mathbb{Q}$ . But for now, we will assume the existence of  $\mathbb{R}$  and its properties.

**Theorem 1.3.1.** If  $x, y \in \mathbb{R}$  and  $x > 0$ , then there exists some  $n \in \mathbb{N}$  such that  $nx > y$ . This theorem is called the **Archimedean property** of  $\mathbb{R}$ .

*Proof.* Suppose  $x, y \in \mathbb{R}$  and  $x > 0$ . Let  $E := \{nx \mid nx \leq y, n \in \mathbb{N}\}$ . If  $E = \emptyset$ , which implies that  $x > y$ ,  $n = 1$  satisfies  $nx > y$ . Otherwise,  $E$  is nonempty and bounded above by  $y$ .

By the least upper bound property, there exists  $\alpha = \sup E \in \mathbb{R}$ . Then  $\alpha - x$  is not an

upper bound of  $E$ . Therefore, there exists some  $m \in \mathbb{N}$  such that  $mx > \alpha - x \implies (m+1)x > \alpha$ . Then  $(m+1)x \notin E$ , which implies that  $(m+1)x > y$ . Therefore, we can take  $n = m+1$ .  $\square$

**Corollary 1.3.2.** *For any  $\epsilon > 0$ , there exists some  $n \in \mathbb{N}$  such that  $b^{-n} < \epsilon$  for any  $b \in \mathbb{R}$  with  $b > 1$ .*

*Proof.* For all  $n \in \mathbb{N}$  and  $x \in \mathbb{R}$  with  $x \geq -1$ , we have  $(1+x)^n \geq 1+nx$ . Let  $a \in \mathbb{R}^+$  be given. By [Theorem 1.3.1](#), there exists some  $n \in \mathbb{N}$  such that  $na > \frac{1}{\epsilon}$ . Then we have

$$(1+a)^n \geq 1+na > na > \frac{1}{\epsilon} \implies (1+a)^{-n} < \epsilon.$$

Since  $b > 1$ , we can take  $a = b-1 > 0$  to complete the proof.  $\square$

*Remark.* [Theorem 1.2.2](#) supposes that  $E$  is bounded in  $\mathbb{Z}$ . By [Theorem 1.3.1](#), we can always find an integer  $k$  such that  $k < x$  or  $x < k$  for any  $x \in \mathbb{R}$ . Therefore, [Theorem 1.2.2](#) also holds when  $E$  is bounded in  $\mathbb{R}$ , which is followed by the fact that  $E$  is also bounded in  $\mathbb{Z}$ .

**Definition 1.3.3.** For any  $x \in \mathbb{R}$ , the **floor function**  $\lfloor x \rfloor$  and **ceiling function**  $\lceil x \rceil$  are defined as follows:

$$\lfloor x \rfloor := \max\{k \in \mathbb{Z} \mid k \leq x\}, \quad \lceil x \rceil := \min\{k \in \mathbb{Z} \mid k \geq x\}$$

**Theorem 1.3.4.** *Let  $x \in \mathbb{R}$  be given. Then  $\lfloor x \rfloor$  and  $\lceil x \rceil$  exist and are unique.*

*Proof.* Let  $E := \{k \in \mathbb{Z} \mid k \leq x\}$ . If  $x \geq 0$ , then  $-1 \in E$ . Otherwise, by [Theorem 1.3.1](#), there exists some  $n \in \mathbb{N}$  such that  $n > -x$ , which implies that  $-n \in E$ . Then, by [Theorem 1.2.2](#),  $E$  has a unique greatest element.

Let  $F := \{k \in \mathbb{Z} \mid k \geq x\}$ . By [Theorem 1.3.1](#), there exists some  $n \in \mathbb{N}$  such that  $n > x$ , which implies that  $n \in F$ . Then, by [Theorem 1.2.2](#),  $F$  has a unique least element.  $\square$

**Theorem 1.3.5.** *If  $x, y \in \mathbb{R}$  and  $x < y$ , then there exists some  $p \in \mathbb{Q}$  such that  $x < p < y$ . This theorem states that  $\mathbb{Q}$  is **dense** in  $\mathbb{R}$ .*

*Proof.* Let  $\epsilon = y - x > 0$ . By [Theorem 1.3.1](#), there exists some  $n \in \mathbb{N}$  such that  $\frac{1}{n} < \epsilon$ . Let  $S := \left\{ k \in \mathbb{Z} \mid \frac{k}{n} < x \right\}$ . Since  $n[x] - 1 \in S$ ,  $S$  is nonempty and bounded above by  $nx$ . By [Theorem 1.2.2](#),  $S$  has a greatest element  $m = \max S$ .

$$x < \frac{m+1}{n} < x + \frac{1}{n} < x + (y - x) = y.$$

Therefore, we can take  $p = \frac{m+1}{n} \in \mathbb{Q}$ . □

**Theorem 1.3.6.** *For each  $x \in \mathbb{R}^+$  and  $n \in \mathbb{N}$ , there exists a unique  $y \in \mathbb{R}^+$  such that  $y^n = x$ . We denote  $y$  by  $\sqrt[n]{x}$  or  $x^{\frac{1}{n}}$ .*

*Proof.* Let  $E := \{t \in \mathbb{R}^+ \mid t^n < x\}$ . Then  $E \neq \emptyset$  since  $2^{-1} \in E$  if  $x \geq 1$ , and  $x^2 \in E$  otherwise. Moreover,  $E$  is bounded above by  $\max\{x, 1\}$ . By the least upper bound property, there exists  $\alpha = \sup E \in \mathbb{R}^+$ . We will show that  $\alpha^n = x$ .

For any  $a, b \in \mathbb{R}^+$  with  $a < b$ , we have

$$b^n - a^n = (b - a) \sum_{k=0}^{n-1} b^{n-1-k} a^k < (b - a) n b^{n-1}.$$

Assume that  $\alpha^n < x$ . By [Theorem 1.3.5](#), we can choose  $h \in \mathbb{R}^+$  such that

$$h < \min \left\{ 1, \frac{x - \alpha^n}{n(\alpha + 1)^{n-1}} \right\}.$$

Put  $a = \alpha$  and  $b = \alpha + h$ . Then we have

$$\begin{aligned} (\alpha + h)^n - \alpha^n &< hn(\alpha + h)^{n-1} < hn(\alpha + 1)^{n-1} \\ \implies (\alpha + h)^n &< \alpha^n + hn(\alpha + 1)^{n-1} < \alpha^n + (x - \alpha^n) = x. \end{aligned}$$

This shows that  $\alpha + h \in E$ , which contradicts the fact that  $\alpha = \sup E$ .

Assume that  $\alpha^n > x$ . By [Theorem 1.3.5](#), we can choose  $h \in \mathbb{R}^+$  such that

$$h < \min \left\{ 1, \alpha, \frac{\alpha^n - x}{n(\alpha + 1)^{n-1}} \right\}.$$

Put  $a = \alpha - h$  and  $b = \alpha$ . Then we have

$$\begin{aligned} \alpha^n - (\alpha - h)^n &< hn\alpha^{n-1} < hn(\alpha + 1)^{n-1} \\ \implies (\alpha - h)^n &> \alpha^n - hn(\alpha + 1)^{n-1} > \alpha^n - (\alpha^n - x) = x. \end{aligned}$$

This shows that  $\alpha - h$  is an upper bound of  $E$ , which contradicts the fact that  $\alpha = \sup E$ .

Therefore, we have  $\alpha^n = x$ , which guarantees the *existence* of  $y$ . For any  $\alpha, \beta \in \mathbb{R}^+$ ,  $\alpha \neq \beta \implies \alpha^n \neq \beta^n$ . Thus the *uniqueness* of  $y$  is guaranteed.  $\square$

**Definition 1.3.7.** Let  $y \in [0, 1)$  and  $b \in \mathbb{N}$  be given with  $b \geq 2$ . A sequence  $\{a_n\}_{n=1}^\infty$  is called the **greedy base  $b$  expansion** of  $y$  if for each  $n \in \mathbb{N}$ , we have

$$0 \leq y - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n}$$

where  $a_k \in S_b := \{0, 1, 2, \dots, b-1\}$  for all  $k \in \mathbb{N}$ . Especially, if  $b = 10$ , then the sequence is called the **greedy decimal expansion** of  $y$ .

**Theorem 1.3.8.** For any  $y \in [0, 1)$ , the greedy base  $b$  expansion of  $y$  exists and is unique.

*Proof.* We define sequences  $\{t_n\}_{n=1}^\infty$ ,  $\{T_n\}_{n=1}^\infty$  and  $\{a_n\}_{n=1}^\infty$  as follows:

- $t_1 = by$ ,  $T_1 := \{t \in S_b \mid t \leq t_1\}$ ,  $a_1 = \max T_1$ .
  - $t_2 = b^2y - ba_1$ ,  $T_2 := \{t \in S_b \mid t \leq t_2\}$ ,  $a_2 = \max T_2$ .
  - $\vdots$
  - $t_n = b^n y - \sum_{k=1}^{n-1} a_k b^{n-k}$ ,  $T_n := \{t \in S_b \mid t \leq t_n\}$ ,  $a_n = \max T_n$
- for all  $n \in \mathbb{N}$  with  $n \geq 2$ ,

Suppose that  $t_n \geq 0$  for some  $n \in \mathbb{N}$ . Since  $0 \in T_n$  and  $T_n$  is bounded above,  $T_n$  has a greatest element by [Theorem 1.2.2](#), which implies that  $\{a_n\}_{n=1}^\infty$  is well-defined. Let  $P(n) : t_n \geq 0$  and  $a_n$  is well-defined.

- **Base case:**  $P(1)$  is true since  $t_1 = by \geq 0$ .
- **Inductive step:** Suppose that  $P(n)$  is true for some  $n \in \mathbb{N}$ . Then we have

$$\begin{aligned} t_{n+1} &= b^{n+1}y - \sum_{k=1}^n a_k b^{n+1-k} \\ &= b \left( b^n y - \sum_{k=1}^n a_k b^{n-k} \right) \\ &= b \left( b^n y - \sum_{k=1}^{n-1} a_k b^{n-k} - a_n \right) \\ &= b(t_n - a_n) \geq 0. \end{aligned}$$

Therefore,  $P(n+1)$  is true whenever  $P(n)$  is true.

By the principle of mathematical induction,  $P(n)$  is true for all  $n \in \mathbb{N}$ . Then, by the definition of  $a_n$ , we have

$$\begin{aligned} a_n &\leq b^n y - \sum_{k=1}^{n-1} a_k b^{n-k} < a_n + 1 \\ \implies 0 &\leq b^n y - \sum_{k=1}^n a_k b^{n-k} < 1 \\ \implies 0 &\leq y - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n} \end{aligned}$$

for each  $n \in \mathbb{N}$ . Therefore, the *existence* of the *greedy base  $b$  expansion* of  $y$  is guaranteed.

Next, suppose that there exist two sequences  $\{a_n\}_{n=1}^{\infty}$  and  $\{c_n\}_{n=1}^{\infty}$  such that for each  $n \in \mathbb{N}$ , we have

- $0 \leq y - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n}$
- $0 \leq y - \sum_{k=1}^n \frac{c_k}{b^k} < \frac{1}{b^n}$

where  $a_k \in S_b = \{0, 1, 2, \dots, b-1\}$  for all  $k = 1, 2, \dots, n$ . We will show that for each  $n \in \mathbb{N}$ , we have  $a_n = c_n$ . Let  $Q(n) : a_k = c_k$  for all  $k = 1, 2, \dots, n$ .

• **Base case:** We have

- $0 \leq y - \frac{a_1}{b} < \frac{1}{b} \implies \frac{a_1}{b} \leq y < \frac{a_1 + 1}{b}.$
- $0 \leq y - \frac{c_1}{b} < \frac{1}{b} \implies \frac{c_1}{b} \leq y < \frac{c_1 + 1}{b}.$

By combining the two inequalities, we have

- $\frac{a_1}{b} \leq y < \frac{c_1 + 1}{b} \implies a_1 < c_1 + 1 \implies a_1 \leq c_1.$
- $\frac{c_1}{b} \leq y < \frac{a_1 + 1}{b} \implies c_1 < a_1 + 1 \implies c_1 \leq a_1.$

Therefore,  $a_1 = c_1$ , which shows that  $Q(1)$  is true.

• **Inductive step:** Suppose that  $Q(n)$  is true for some  $n \in \mathbb{N}$ . We have

- $0 \leq y - \sum_{k=1}^{n+1} \frac{a_k}{b^k} < \frac{1}{b^{n+1}} \implies \sum_{k=1}^{n+1} \frac{a_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{a_k}{b^k} + \frac{1}{b^{n+1}}.$

$$- 0 \leq y - \sum_{k=1}^{n+1} \frac{c_k}{b^k} < \frac{1}{b^{n+1}} \implies \sum_{k=1}^{n+1} \frac{c_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{c_k}{b^k} + \frac{1}{b^{n+1}}.$$

By combining the two inequalities, we have

$$\begin{aligned} - \sum_{k=1}^{n+1} \frac{a_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{c_k}{b^k} + \frac{1}{b^{n+1}} &\implies \sum_{k=1}^{n+1} \frac{a_k - c_k}{b^k} < \frac{1}{b^{n+1}}. \\ - \sum_{k=1}^{n+1} \frac{c_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{a_k}{b^k} + \frac{1}{b^{n+1}} &\implies \sum_{k=1}^{n+1} \frac{c_k - a_k}{b^k} < \frac{1}{b^{n+1}}. \end{aligned}$$

By the inductive hypothesis, we have  $a_k = c_k$  for all  $k = 1, 2, \dots, n$ . Therefore, we have

$$\begin{aligned} - \frac{a_{n+1} - c_{n+1}}{b^{n+1}} < \frac{1}{b^{n+1}} &\implies a_{n+1} - c_{n+1} < 1 \implies a_{n+1} \leq c_{n+1}. \\ - \frac{c_{n+1} - a_{n+1}}{b^{n+1}} < \frac{1}{b^{n+1}} &\implies c_{n+1} - a_{n+1} < 1 \implies c_{n+1} \leq a_{n+1}. \end{aligned}$$

Therefore,  $a_{n+1} = c_{n+1}$ , and thus  $Q(n+1)$  is true whenever  $Q(n)$  is true.

By the principle of mathematical induction,  $Q(n)$  is true for all  $n \in \mathbb{N}$ . Therefore, the *uniqueness* of the *greedy base  $b$  expansion* of  $y$  is guaranteed.  $\square$

**Theorem 1.3.9.** *Let  $\{a_n\}_{n=1}^{\infty}$  be the greedy base  $b$  expansion of  $a \in [0, 1)$ , and let  $\{c_n\}_{n=1}^{\infty}$  be the greedy base  $b$  expansion of  $c \in [0, 1)$  for some  $b \in \mathbb{N}$  with  $b \geq 2$ . If  $a_n = c_n$  for all  $n \in \mathbb{N}$ , then  $a = c$ .*

*Proof.* Suppose that  $a \neq c$ . Without loss of generality, assume that  $a < c$ . Then, by [Corollary 1.3.2](#), there exists some  $n \in \mathbb{N}$  such that

$$b^n > \frac{1}{c - a} \implies \frac{1}{b^n} < c - a.$$

By the definition of the *greedy base  $b$  expansion*, we have

$$\begin{aligned} \bullet \quad 0 \leq a - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n} &\implies \sum_{k=1}^n \frac{a_k}{b^k} + \frac{1}{b^n} \leq a + \frac{1}{b^n}. \\ \bullet \quad 0 \leq c - \sum_{k=1}^n \frac{c_k}{b^k} < \frac{1}{b^n} &\implies c < \sum_{k=1}^n \frac{c_k}{b^k} + \frac{1}{b^n}. \end{aligned}$$

By the assumption that  $a_n = c_n$  for all  $n \in \mathbb{N}$ , we have

$$\sum_{k=1}^n \frac{a_k}{b^k} = \sum_{k=1}^n \frac{c_k}{b^k}.$$

Therefore, we have a contradiction:

$$\sum_{k=1}^n \frac{a_k}{b^k} + \frac{1}{b^n} \leq a + \frac{1}{b^n} < c < \sum_{k=1}^n \frac{c_k}{b^k} + \frac{1}{b^n}.$$

□

**Theorem 1.3.10.** *Let  $b \in \mathbb{N}$  with  $b \geq 2$  and  $\{a_n\}_{n=1}^\infty$  be given with  $a_n \in S_b := \{0, 1, 2, \dots, b-1\}$  for all  $n \in \mathbb{N}$ . If there exists no  $y \in [0, 1)$  such that  $\{a_n\}_{n=1}^\infty$  is the greedy base  $b$  expansion of  $y$ , then there exists some  $N \in \mathbb{N}$  such that*

$$a_n = b - 1 \quad \text{for all } n \geq N.$$

*Proof.* We will prove the contrapositive of the statement: For all  $N \in \mathbb{N}$ , if there exists some  $n \in \mathbb{N}$  such that

$$n \geq N \text{ and } a_n \neq b - 1,$$

then there exists  $y \in [0, 1)$  such that  $\{a_n\}_{n=1}^\infty$  is the greedy base  $b$  expansion of  $y$ .

Assume that the condition of the above claim holds. Let  $x_n = \sum_{k=1}^n \frac{a_k}{b^k}$  for each  $n \in \mathbb{N}$ . Since  $a_k \in S_b$  for all  $k \in \mathbb{N}$ , we have

$$0 \leq x_n \leq \sum_{k=1}^n \frac{b-1}{b^k} = 1 - \frac{1}{b^n} < 1 \quad \text{for all } n \in \mathbb{N}.$$

Then  $X := \{x_n \mid n \in \mathbb{N}\}$  is bounded above. By the [least upper bound property](#), there exists  $y = \sup X \geq 0$ . First, we will show that  $y < 1$ . By the assumption, there exists  $m \in \mathbb{N}$  such that  $a_m \neq b - 1$ . Then, for all  $n > m$ , we have

$$\begin{aligned} x_n &= x_m + \sum_{k=m+1}^n \frac{a_k}{b^k} \leq x_m + \sum_{k=m+1}^n \frac{b-1}{b^k} \\ \implies x_n &\leq x_m + \frac{1}{b^m} - \frac{1}{b^n} < x_m + \frac{1}{b^m}. \end{aligned}$$

Since  $a_m \leq b - 2$ , we have

$$\begin{aligned} x_m + \frac{1}{b^m} &= \sum_{k=1}^{m-1} \frac{a_k}{b^k} + \frac{a_m + 1}{b^m} \\ &\leq \sum_{k=1}^{m-1} \frac{a_k}{b^k} + \frac{b-1}{b^m} \\ &\leq \sum_{k=1}^m \frac{b-1}{b^k} = 1 - \frac{1}{b^m}. \end{aligned}$$

We have  $x_n \leq x_m$  for all  $n \leq m$ , and  $x_n < x_m + \frac{1}{b^m} \leq 1 - \frac{1}{b^m}$  for all  $n > m$ . Thus,  $1 - \frac{1}{b^m}$  is an upper bound of  $X$ . Since  $y = \sup X$ , we have

$$y \leq 1 - \frac{1}{b^m} < 1 \implies y \in [0, 1).$$

Next, we will show that  $\{a_n\}_{n=1}^\infty$  is the *greedy base  $b$  expansion* of  $y$ . Let  $n \in \mathbb{N}$  be fixed. By the assumption, there exists  $m \in \mathbb{N}$  such that  $m \geq n$  and  $a_m \leq b - 2$ .

$$\begin{aligned} x_m &= x_n + \sum_{k=n+1}^m \frac{a_k}{b^k} < x_n + \sum_{k=n+1}^m \frac{b-1}{b^k} \\ \implies x_m &< x_n + \frac{1}{b^n} - \frac{1}{b^m} \implies x_m + \frac{1}{b^m} < x_n + \frac{1}{b^n}. \end{aligned}$$

Since  $a_m \leq b - 2$ ,  $x_m + \frac{1}{b^m}$  is an upper bound of  $X$  by the same argument as above. Since  $y = \sup X$ , we have

$$y \leq x_m + \frac{1}{b^m} < x_n + \frac{1}{b^n} \implies 0 \leq y - x_n < \frac{1}{b^n}.$$

Thus,  $\{a_n\}_{n=1}^\infty$  is the *greedy base  $b$  expansion* of  $y$ . □

**Theorem 1.3.11.** *Let  $b \in \mathbb{N}$  with  $b \geq 2$  and  $\{a_n\}_{n=1}^\infty$  be given with  $a_n \in S_b := \{0, 1, 2, \dots, b-1\}$  for all  $n \in \mathbb{N}$ . If there exists some  $N \in \mathbb{N}$  such that*

$$a_n = b - 1 \quad \text{for all } n \geq N,$$

*then there exists no  $y \in [0, 1)$  such that  $\{a_n\}_{n=1}^\infty$  is the greedy base  $b$  expansion of  $y$ . This theorem is the converse of [Theorem 1.3.10](#).*

*Proof.* Suppose that there exists some  $y \in [0, 1)$  such that  $\{a_n\}_{n=1}^\infty$  is the *greedy base  $b$  expansion* of  $y$ . Then we have

$$0 \leq y - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n} \quad \text{for all } n \in \mathbb{N}.$$

Let  $N \in \mathbb{N}$  be such that  $a_n = b - 1$  for all  $n \geq N$ . Let  $x_n = \sum_{k=1}^n \frac{a_k}{b^k}$  for each  $n \in \mathbb{N}$ .

Then, for all  $n \geq N$ , we have

$$\begin{aligned} x_n &= x_{N-1} + \sum_{k=N}^n \frac{b-1}{b^k} = x_{N-1} + \frac{1}{b^{N-1}} - \frac{1}{b^n} \\ \implies y - x_n &= y - x_{N-1} - \frac{1}{b^{N-1}} + \frac{1}{b^n} \geq 0 \\ \implies \frac{1}{b^n} &\geq x_{N-1} + \frac{1}{b^{N-1}} - y. \end{aligned}$$

Since  $0 \leq y - x_{N-1} < \frac{1}{b^{N-1}}$ , we have  $x_{N-1} + \frac{1}{b^{N-1}} - y > 0$ . Then, by [Corollary 1.3.2](#), there exists some  $n \in \mathbb{N}$  such that

$$\frac{1}{b^n} < x_{N-1} + \frac{1}{b^{N-1}} - y,$$

which is a contradiction. Therefore, there exists no  $y \in [0, 1)$  such that  $\{a_n\}_{n=1}^\infty$  is the greedy base  $b$  expansion of  $y$ .  $\square$

## 1.4 The Extended Real Number System

The Extended real number system  $\overline{\mathbb{R}}$  is defined as

$$\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}.$$

We define the following operations on  $\overline{\mathbb{R}}$ :

- $x + (+\infty) = +\infty$  for all  $x \in \mathbb{R} \cup \{\infty\}$ .
- $x + (-\infty) = -\infty$  for all  $x \in \mathbb{R} \cup \{-\infty\}$ .
- $x \cdot (\pm\infty) = \pm\infty$  for all  $x \in \mathbb{R}^+ \cup \{\infty\}$ .
- $x \cdot (\pm\infty) = \mp\infty$  for all  $x \in \mathbb{R}^- \cup \{-\infty\}$ .
- $\frac{x}{\pm\infty} = 0$  for all  $x \in \mathbb{R}$ .

$\overline{\mathbb{R}}$  is not a field, but it is an ordered set with the following order relation:

$$-\infty < x < +\infty \quad \text{for all } x \in \mathbb{R}.$$

**Theorem 1.4.1.** *Every subset of  $\overline{\mathbb{R}}$  has a least upper bound and a greatest lower bound in  $\overline{\mathbb{R}}$ .*

*Proof.* Let  $E \subseteq \overline{\mathbb{R}}$ . If  $E = \emptyset$ , then  $\sup E = -\infty$  and  $\inf E = +\infty$ . If  $E \neq \emptyset$ , then we have the following cases:

- If  $E$  is bounded above in  $\mathbb{R}$ , then  $\sup E$  exists in  $\mathbb{R}$  by the least upper bound property of  $\mathbb{R}$ .
- If  $E$  is not bounded above in  $\mathbb{R}$ , then  $\sup E = +\infty$ .
- If  $E$  is bounded below in  $\mathbb{R}$ , then  $\inf E$  exists in  $\mathbb{R}$  by the greatest lower bound property of  $\mathbb{R}$ .
- If  $E$  is not bounded below in  $\mathbb{R}$ , then  $\inf E = -\infty$ .

$\square$

## 1.5 The Complex Field

The complex field  $\mathbb{C}$  is defined as

$$\mathbb{C} := \{(x, y) \mid x, y \in \mathbb{R}\}.$$

We define the following operations on  $\mathbb{C}$ :

- $(x, y) + (u, v) = (x + u, y + v)$  for all  $(x, y), (u, v) \in \mathbb{C}$ .
- $(x, y) \cdot (u, v) = (xu - yv, xv + yu)$  for all  $(x, y), (u, v) \in \mathbb{C}$ .

$\mathbb{C}$  is a field with the following properties:

- The additive identity is  $(0, 0)$  and the multiplicative identity is  $(1, 0)$ .
- The additive inverse of  $(x, y)$  is  $(-x, -y)$  and the multiplicative inverse of  $(x, y)$  is

$$\left( \frac{x}{x^2 + y^2}, -\frac{y}{x^2 + y^2} \right)$$

for all  $(x, y) \in \mathbb{C}$  with  $(x, y) \neq (0, 0)$ .

*Remark.* The function

$$f : \{(x, 0) \mid x \in \mathbb{R}\} \subseteq \mathbb{C} \rightarrow \mathbb{R}$$

defined by  $f((x, 0)) = x$  is an *isomorphism*. Then we can identify  $\mathbb{R}$  with the subfield  $\{(x, 0) \mid x \in \mathbb{R}\}$  of  $\mathbb{C}$ . Now we will denote  $(x, 0) \in \mathbb{C}$  by  $x \in \mathbb{R}$ .

*Remark.* We define  $i = (0, 1) \in \mathbb{C}$ . Then we have  $i^2 = -1$ , and any complex number  $(x, y) \in \mathbb{C}$  can be written as

$$(x, y) = x + yi.$$

**Definition 1.5.1.** Let  $z = x + yi \in \mathbb{C}$  for some  $x, y \in \mathbb{R}$ . The **real part** of  $z$  is defined as

$$\Re(z) = x \text{ or } \operatorname{Re}(z) = x,$$

the **imaginary part** of  $z$  is defined as

$$\Im(z) = y \text{ or } \operatorname{Im}(z) = y,$$

and the **complex conjugate** of  $z$  is defined as

$$\bar{z} = x - yi.$$

**Theorem 1.5.2.** *Let  $z, w \in \mathbb{C}$ . Then we have the following properties:*

- $\overline{(\overline{z})} = z$ .
- $z = \overline{z} \iff z \in \mathbb{R}$ .
- $\overline{z + w} = \overline{z} + \overline{w}$ .
- $\overline{zw} = \overline{z} \cdot \overline{w}$ .
- $z + \overline{z} = 2\operatorname{Re}(z)$  and  $z - \overline{z} = 2i\operatorname{Im}(z)$ .
- $z\overline{z} = x^2 + y^2 \in \mathbb{R}^+ \cup \{0\}$ .

*Proof.* The proof follows directly from the definitions and properties of the operations on  $\mathbb{C}$ .  $\square$

**Definition 1.5.3.** Let  $z \in \mathbb{C}$ . the **modulus** of  $z$  is defined as

$$|z| = \sqrt{z\overline{z}} = \sqrt{x^2 + y^2} \in \mathbb{R}^+ \cup \{0\}.$$

The *existence* and *uniqueness* of  $|z|$  follows from the last part of [Theorem 1.5.2](#) and [Theorem 1.3.6](#).

**Theorem 1.5.4.** *Let  $a_1, a_2, \dots, a_n \in \mathbb{C}$  and  $b_1, b_2, \dots, b_n \in \mathbb{C}$ . Then we have*

$$\left| \sum_{k=1}^n a_k \overline{b_k} \right|^2 \leq \sum_{k=1}^n |a_k|^2 \cdot \sum_{k=1}^n |b_k|^2.$$

*This theorem is called the **Cauchy-Schwarz inequality**.*

*Proof.* Let  $i, j \in \mathbb{N}$  with  $1 \leq i < j \leq n$ . Then we have

$$\begin{aligned} |a_i b_j - a_j b_i|^2 &= (a_i b_j - a_j b_i) \overline{(a_i b_j - a_j b_i)} \\ &= |a_i|^2 |b_j|^2 + |a_j|^2 |b_i|^2 - (a_i \overline{b_i} \cdot \overline{a_j} b_j + \overline{a_i} b_i \cdot a_j \overline{b_j}). \end{aligned}$$

Consider the following sums:

$$\begin{aligned} \sum_{i < j} (|a_i|^2 |b_j|^2 + |a_j|^2 |b_i|^2) &= \sum_{i < j} (|a_i|^2 |b_j|^2) + \sum_{j < i} (|a_i|^2 |b_j|^2) \\ &= \sum_{i \neq j} (|a_i|^2 |b_j|^2) \\ &= \sum_{k=1}^n |a_k|^2 \cdot \sum_{k=1}^n |b_k|^2 - \sum_{k=1}^n |a_k|^2 |b_k|^2. \end{aligned}$$

$$\begin{aligned}
\sum_{i < j} (a_i \bar{b}_i \cdot \bar{a}_j b_j + \bar{a}_i b_i \cdot a_j \bar{b}_j) &= \sum_{i \neq j} (a_i \bar{b}_i \cdot \bar{a}_j b_j) \\
&= \sum_{i=1}^n a_i \bar{b}_i \cdot \sum_{j=1}^n \bar{a}_j b_j - \sum_{k=1}^n a_k \bar{b}_k \cdot \bar{a}_k b_k \\
&= \left| \sum_{k=1}^n a_k \bar{b}_k \right|^2 - \sum_{k=1}^n |a_k|^2 |b_k|^2.
\end{aligned}$$

Therefore, we have

$$\sum_{i < j} |a_i b_j - a_j b_i|^2 = \sum_{k=1}^n |a_k|^2 \cdot \sum_{k=1}^n |b_k|^2 - \left| \sum_{k=1}^n a_k \bar{b}_k \right|^2 \geq 0.$$

□

**Corollary 1.5.5.** Let  $a_1, a_2, \dots, a_n \in \mathbb{R}$  and  $b_1, b_2, \dots, b_n \in \mathbb{R}$ . Then we have

$$\left( \sum_{k=1}^n a_k b_k \right)^2 \leq \sum_{k=1}^n a_k^2 \cdot \sum_{k=1}^n b_k^2.$$

**Theorem 1.5.6.** Let  $z, w \in \mathbb{C}$ . Then we have the following properties:

- (A)  $|zw| = |z| \cdot |w|$ .
- (B)  $|z + w| \leq |z| + |w|$ .
- (C)  $||z| - |w|| \leq |z - w|$ .

*Proof.* Let  $z = x + yi$  and  $w = u + vi$  for some  $x, y, u, v \in \mathbb{R}$ . Then we have

- (A)  $|zw|^2 = (xu - yv)^2 + (xv + yu)^2 = (x^2 + y^2)(u^2 + v^2) = |z|^2 \cdot |w|^2$ .
- (B) By [Corollary 1.5.5](#), we have  $(xu + yv)^2 \leq (x^2 + y^2)(u^2 + v^2)$ .

$$\begin{aligned}
|z + w|^2 &= (x + u)^2 + (y + v)^2 \\
&= x^2 + y^2 + u^2 + v^2 + 2(xu + yv) \\
&\leq x^2 + y^2 + u^2 + v^2 + 2\sqrt{x^2 + y^2} \cdot \sqrt{u^2 + v^2} \\
&= (|z| + |w|)^2.
\end{aligned}$$

- (C) Again, by [Corollary 1.5.5](#), we have

$$\begin{aligned}
|z - w|^2 &= (x - u)^2 + (y - v)^2 \\
&= x^2 + y^2 + u^2 + v^2 - 2(xu + yv) \\
&\geq x^2 + y^2 + u^2 + v^2 - 2\sqrt{x^2 + y^2} \cdot \sqrt{u^2 + v^2} \\
&= (|z| - |w|)^2.
\end{aligned}$$

□

## 1.6 Euclidean Spaces

For  $n \in \mathbb{N}$ , we define the **Euclidean  $n$ -space**  $\mathbb{R}^n$  as the set of all ordered  $n$ -tuples of real numbers, i.e.,

$$\mathbb{R}^n := \{ \mathbf{x} = (x_1, x_2, \dots, x_n) \mid x_k \in \mathbb{R} \text{ for all } k = 1, 2, \dots, n \}$$

where  $x_1, x_2, \dots, x_n$  are called the **coordinates** of  $\mathbf{x}$ . The elements of  $\mathbb{R}^n$  with  $n > 1$  are called **vectors** or *points*, which are denoted by boldface letters.

Let  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$  and  $\alpha \in \mathbb{R}$ . We define the following operations on  $\mathbb{R}^n$ :

- $\mathbf{x} + \mathbf{y} = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$ .
- $\alpha \mathbf{x} = (\alpha x_1, \alpha x_2, \dots, \alpha x_n)$ .

Then  $\mathbb{R}^n$  is a *vector space* over  $\mathbb{R}$  with the zero vector  $\mathbf{0} = (0, 0, \dots, 0)$ .

**Definition 1.6.1.** Let  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ . The **inner product** of  $\mathbf{x}$  and  $\mathbf{y}$  is defined as

$$\mathbf{x} \cdot \mathbf{y} = \sum_{k=1}^n x_k y_k.$$

The **norm** of  $\mathbf{x}$  is defined as

$$\|\mathbf{x}\| = \sqrt{\mathbf{x} \cdot \mathbf{x}} = \sqrt{\sum_{k=1}^n x_k^2}.$$

The **distance** between  $\mathbf{x}$  and  $\mathbf{y}$  is defined as

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\| = \sqrt{\sum_{k=1}^n (x_k - y_k)^2}.$$

We can confirm that  $d(x, y) = |x - y|$  in the case of  $n = 1$ .

**Theorem 1.6.2.** Let  $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$  and  $\alpha \in \mathbb{R}$ . Then we have the following properties:

- (A)  $|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \cdot \|\mathbf{y}\|$ .
- (B)  $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ .
- (C)  $\|\mathbf{x} - \mathbf{y}\| \geq \|\mathbf{x}\| - \|\mathbf{y}\|$ .

*Proof.* (A) By [Corollary 1.5.5](#), we have

$$|\mathbf{x} \cdot \mathbf{y}|^2 = \left| \sum_{k=1}^n x_k y_k \right|^2 \leq \sum_{k=1}^n x_k^2 \cdot \sum_{k=1}^n y_k^2 = \|\mathbf{x}\|^2 \cdot \|\mathbf{y}\|^2.$$

(B) By part (A), we have

$$\begin{aligned} \|\mathbf{x} + \mathbf{y}\|^2 &= (\mathbf{x} + \mathbf{y}) \cdot (\mathbf{x} + \mathbf{y}) = \|\mathbf{x}\|^2 + 2\mathbf{x} \cdot \mathbf{y} + \|\mathbf{y}\|^2 \\ &\leq \|\mathbf{x}\|^2 + 2|\mathbf{x} \cdot \mathbf{y}| + \|\mathbf{y}\|^2 \leq \|\mathbf{x}\|^2 + 2\|\mathbf{x}\| \cdot \|\mathbf{y}\| + \|\mathbf{y}\|^2 \\ &= (\|\mathbf{x}\| + \|\mathbf{y}\|)^2. \end{aligned}$$

(C) By part (B), we have

$$\begin{aligned} \|\mathbf{x}\| &= \|(\mathbf{x} - \mathbf{y}) + \mathbf{y}\| \leq \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y}\| \implies \|\mathbf{x}\| - \|\mathbf{y}\| \leq \|\mathbf{x} - \mathbf{y}\|. \\ \|\mathbf{y}\| &= \|(\mathbf{y} - \mathbf{x}) + \mathbf{x}\| \leq \|\mathbf{y} - \mathbf{x}\| + \|\mathbf{x}\| \implies \|\mathbf{y}\| - \|\mathbf{x}\| \leq \|\mathbf{x} - \mathbf{y}\|. \end{aligned}$$

□

## Chapter 2

# Basic Topology

### 2.1 Relations

A **relation**  $\sim$  on a set  $S$  defines a subset  $R$  of  $S \times S$ , and vice versa.

$$R := \{(x, y) \in S \times S \mid x \sim y\}, \quad x \sim y \iff (x, y) \in R.$$

**Definition 2.1.1.** A relation  $\sim$  on a set  $S$  is called an **equivalence relation** if it satisfies the following properties:

- Reflexivity:  $x \sim x$  for all  $x \in S$ .
- Symmetry: If  $x \sim y$ , then  $y \sim x$  for all  $x, y \in S$ .
- Transitivity: If  $x \sim y$  and  $y \sim z$ , then  $x \sim z$  for all  $x, y, z \in S$ .

**Definition 2.1.2.** A **partition** of a set  $S$  is a collection  $\{A_\alpha\}_{\alpha \in I}$  of nonempty subsets of  $S$  such that

- $A_\alpha \cap A_\beta = \emptyset$  for all  $\alpha, \beta \in I$  with  $\alpha \neq \beta$ .
- $\bigcup_{\alpha \in I} A_\alpha = S$ .

**Definition 2.1.3.** Let  $\sim$  be an equivalence relation on a set  $S$ . For each  $x \in S$ , the **equivalence class** of  $x$  is defined as

$$[x] := \{y \in S \mid y \sim x\}.$$

We denote the set of all equivalence classes of  $S$  by  $S/\sim := \{[x] \mid x \in S\}$ .

**Theorem 2.1.4.** *Let  $\sim$  be an equivalence relation on a set  $S$ . For any  $x, y \in S$ , we have either*

$$[x] = [y] \quad \text{or} \quad [x] \cap [y] = \emptyset.$$

*Therefore,  $S/\sim$  is a partition of  $S$ .*

*Proof.* Suppose  $[x] \cap [y] \neq \emptyset$ . Then there exists  $z \in S$  such that  $z \in [x]$  and  $z \in [y]$ . This implies that  $z \sim x$  and  $z \sim y$ . By the symmetry and transitivity of  $\sim$ , we have  $x \sim z$  and  $z \sim y$ , which implies that  $x \sim y$ .

Therefore, for any  $w \in [x]$ , we have  $w \sim x$  and  $x \sim y$ , which implies that  $w \sim y$ , so  $w \in [y]$ . This shows that  $[x] \subseteq [y]$ . Similarly, we can show that  $[y] \subseteq [x]$ , so  $[x] = [y]$ .  $\square$

**Theorem 2.1.5.** *Let  $\{A_\alpha\}_{\alpha \in I}$  be a partition of a set  $S$ . Define a relation  $\sim$  on  $S$  by*

$$x \sim y \iff x, y \in A_\alpha \text{ for some } \alpha \in I.$$

*Then  $\sim$  is an equivalence relation on  $S$ , and the equivalence classes of  $\sim$  are precisely the sets  $A_\alpha$ .*

*Proof.* We need to show that  $\sim$  is reflexive, symmetric, and transitive.

- Reflexivity: For any  $x \in S$ , there exists  $\alpha \in I$  such that  $x \in A_\alpha$ , so  $x \sim x$ .
- Symmetry: If  $x \sim y$ , then there exists  $\alpha \in I$  such that  $x, y \in A_\alpha$ , so  $y \sim x$ .
- Transitivity: If  $x \sim y$  and  $y \sim z$ , then there exist  $\alpha, \beta \in I$  such that  $x, y \in A_\alpha$  and  $y, z \in A_\beta$ . Since  $y \in A_\alpha \cap A_\beta$  and the sets  $\{A_\alpha\}_{\alpha \in I}$  are disjoint, we must have  $\alpha = \beta$ , so  $x, z \in A_\alpha$  and  $x \sim z$ .

Therefore,  $\sim$  is an equivalence relation on  $S$ . The equivalence class of any  $x \in S$  is precisely the set  $A_\alpha$  containing  $x$ , so the equivalence classes of  $\sim$  are exactly the sets  $A_\alpha$ .  $\square$

## 2.2 Functions

A *function* (or a *mapping*) is defined with 3 components.

- A **domain**  $X$ .
- A **codomain**  $Y$ .
- A **rule**  $f$  which associates each element  $x \in X$  to a *unique* element  $y \in Y$ .

$f$  is said to be a *function* of  $X$  into  $Y$ , which is denoted by  $f : X \rightarrow Y$ . For each  $x \in X$ , the *unique* element  $y \in Y$  associated with  $x$  is denoted by  $f(x)$ .

*Remark.* Let  $f : X \rightarrow Y$  be a function. For any subset  $E \subseteq X$ , we define the **image** of  $E$  under  $f$  as

$$f(E) := \{f(x) \mid x \in E\} \subseteq Y.$$

Especially, if  $E = X$ , then  $f(X) \subseteq Y$  is called the **range** of  $f$ .

*Remark.* Let  $f : X \rightarrow Y$  be a function. For any subset  $E \subseteq Y$ , we define the **inverse image** of  $E$  under  $f$  as

$$f^{-1}(E) := \{x \in X \mid f(x) \in E\} \subseteq X.$$

**Definition 2.2.1.** Let  $f : X \rightarrow Y$  be a function. We define the properties of  $f$  as follows:

- $f$  is **injective** (or *one-to-one*) if for any  $x_1, x_2 \in X$ ,

$$f(x_1) = f(x_2) \implies x_1 = x_2.$$

- $f$  is **surjective** (or *onto*) if  $f(X) = Y$ .
- $f$  is **bijective** (or a *one-to-one correspondence*) if it is both injective and surjective.

**Definition 2.2.2.** Let  $f : X \rightarrow Y$  be a bijective function. Then for any  $y \in Y$ , there exists a *unique*  $x \in X$  such that  $f(x) = y$ . Therefore, we can define a function  $f^{-1} : Y \rightarrow X$  by  $f^{-1}(y) = x$ .  $f^{-1}$  is called the **inverse function** of  $f$ .

**Theorem 2.2.3.** Let  $f : X \rightarrow Y$  be a bijective function. Then the inverse function  $f^{-1} : Y \rightarrow X$  is bijective.

*Proof.* Let  $y_1, y_2 \in Y$ . Suppose that  $f^{-1}(y_1) = f^{-1}(y_2) = x$ . Then, by [Definition 2.2.2](#), we have both  $f(x) = y_1$  and  $f(x) = y_2$ . Since  $f$  is a function, we have  $y_1 = y_2$ , which implies that  $f^{-1}$  is injective.

By the definition of inverse image, we have  $f^{-1}(Y) \subseteq X$ . Also, for any  $x \in X$ , we have  $f(x) \in Y$ , which implies that  $x \in f^{-1}(Y)$ . Therefore, we have  $f^{-1}(Y) \supseteq X \implies f^{-1}(Y) = X$ , which implies that  $f^{-1}$  is surjective.  $\square$

**Definition 2.2.4.** Let  $f : X \rightarrow Y$  be a function. Let  $E \subseteq X$  and  $F \subseteq Y$  such that  $f(E) \subseteq F$ . Then we define the restriction functions as follows:

- The **restriction** of  $f$  to  $E$  is defined by

$$f|_E : E \rightarrow Y, \quad f|_E(x) = f(x) \quad \text{for all } x \in E.$$

- The **Corestriction** of  $f$  to  $F$  is defined by

$$f|_F : X \rightarrow F, \quad f|_F(x) = f(x) \quad \text{for all } x \in X.$$

- The **Bi-restriction** of  $f$  to  $E$  and  $F$  is defined by

$$f|_E^F : E \rightarrow F, \quad f|_E^F(x) = f(x) \quad \text{for all } x \in E.$$

We will simply call all of them as just *restriction*, if there is no ambiguity.

**Theorem 2.2.5.** *Let  $f : X \rightarrow Y$  be a function. Let  $E \subseteq X$  and  $F \subseteq Y$  such that  $f(E) \subseteq F$ . Then we have the following properties:*

- (A) *If  $f$  is injective, then  $f|_E^F$  is also injective.*
- (B) *If  $f(E) = F$ , then  $f|_E^F$  is surjective.*
- (C) *If  $f$  is injective and  $f(E) = F$ , then  $f|_E^F$  is bijective.*

*Proof.* (C) follows from (A) and (B). Let  $x_1, x_2 \in E$ .

- (A) Assume that  $f$  is injective. Then we have

$$f|_E^F(x_1) = f|_E^F(x_2) \implies f(x_1) = f(x_2) \implies x_1 = x_2.$$

- (B) Assume that  $f(E) = F$ . Then for any  $y \in F$ , there exists  $x \in E$  such that  $f(x) = y$ . Therefore, we have  $f|_E^F(E) = F$ .

□

**Definition 2.2.6.** Let  $X, Y, Z$  be sets and  $f : X \rightarrow Y, g : Y \rightarrow Z$  be functions. Then we define the **composition** of  $f$  and  $g$  as the function  $g \circ f : X \rightarrow Z$  defined by

$$(g \circ f)(x) = g(f(x)) \quad \text{for all } x \in X.$$

**Theorem 2.2.7.** *Let  $X, Y, Z$  be sets and  $f : X \rightarrow Y, g : Y \rightarrow Z$  be functions. Then we have the following properties:*

- (A) *If  $f$  and  $g$  are injective, then  $g \circ f$  is also injective.*

(B) If  $f$  and  $g$  are surjective, then  $g \circ f$  is also surjective.

(C) If  $f$  and  $g$  are bijective, then  $g \circ f$  is also bijective.

*Proof.* (C) follows from (A) and (B). Let  $x_1, x_2 \in X$ .

(A) Assume that  $f$  and  $g$  are injective. Then we have

$$\begin{aligned} (g \circ f)(x_1) = (g \circ f)(x_2) &\implies g(f(x_1)) = g(f(x_2)) \\ &\implies f(x_1) = f(x_2) \implies x_1 = x_2. \end{aligned}$$

(B) Assume that  $f$  and  $g$  are surjective. Then for any  $z \in Z$ , there exists  $y \in Y$  such that  $g(y) = z$ . Since  $f$  is surjective, there exists  $x \in X$  such that  $f(x) = y$ . Therefore, we have  $(g \circ f)(x) = g(f(x)) = g(y) = z$ .

□

**Theorem 2.2.8.** Let  $S$  be a set of sets. For [Sections 2.3](#) and [2.4](#), we consider the relation  $\sim$  defined on  $S$  by

$$X \sim Y \iff \text{There exists a bijection } f : X \rightarrow Y.$$

The relation  $\sim$  is an equivalence relation on  $S$ .

*Proof.* Let  $X, Y, Z \in S$ .

- **Reflexivity:** The identity function  $id_X : X \rightarrow X$  defined by  $id_X(x) = x$  for all  $x \in X$  is bijective. Therefore, we have  $X \sim X$ .
- **Symmetry:** Suppose that  $X \sim Y$ . Then there exists a bijection  $f : X \rightarrow Y$ . By [Theorem 2.2.3](#), the inverse function  $f^{-1} : Y \rightarrow X$  is also bijective. Therefore, we have  $Y \sim X$ .
- **Transitivity:** Suppose that  $X \sim Y$  and  $Y \sim Z$ . Then there exist bijections  $f : X \rightarrow Y$  and  $g : Y \rightarrow Z$ . By [Theorem 2.2.7](#), the composition  $g \circ f : X \rightarrow Z$  is also bijective. Therefore, we have  $X \sim Z$ .

□

## 2.3 Finite Sets

Let  $J_n := \{1, 2, \dots, n\}$  for each  $n \in \mathbb{N}$ . A set  $X$  is said to be **finite** if  $J_n \sim X$  for some  $n \in \mathbb{N}$ . In this case,  $n$  is called the **cardinality** of  $X$ , denoted by  $|X|$ . Especially, if  $X = \emptyset$ , we define  $|X| = 0$ .  $X$  is said to be **infinite** if  $X$  is not finite.

**Lemma 2.3.1.** Every subset of  $J_n$  is finite for each  $n \in \mathbb{N}$ .

*Proof.* We will prove the lemma by mathematical induction on  $n$ . Let  $P(n)$  : Every subset of  $J_n$  is finite.

- **Base case:** Let  $n = 1$ . Then  $J_1 = \{1\}$ , whose only subsets are  $\emptyset$  and  $J_1$ .  $J_1 \sim J_1$  by  $f := id_{J_1}$ , and  $\emptyset$  is finite by definition. Thus  $P(1)$  is true.
- **Inductive step:** Assume that  $P(n)$  is true for some  $n \in \mathbb{N}$ . Let  $A \subseteq J_{n+1}$ . We have two cases:
  - If  $n + 1 \notin A$ , then  $A \subseteq J_n$ , which is finite by the inductive hypothesis.
  - If  $n + 1 \in A$ , then  $A = B \cup \{n + 1\}$  for some  $B \subseteq J_n$ . By the inductive hypothesis,  $B$  is finite, so there exists a bijection  $f : J_m \rightarrow B$  for some  $m \in \mathbb{N}$ . Let  $g : J_{m+1} \rightarrow A$  be defined by

$$g(k) = \begin{cases} f(k) & \text{if } k \leq m \\ n + 1 & \text{if } k = m + 1 \end{cases}.$$

Then  $g$  is bijective, which implies that  $A$  is finite.

Thus  $P(n + 1)$  is true whenever  $P(n)$  is true.

By the principle of mathematical induction,  $P(n)$  is true for all  $n \in \mathbb{N}$ . □

**Theorem 2.3.2.** *Every subset of a finite set is finite.*

*Proof.* Let  $X$  be a finite set and  $Y \subseteq X$ . Then there exists a bijection  $f : J_n \rightarrow X$  for some  $n \in \mathbb{N}$ . Let  $Z = f^{-1}(Y) \subseteq J_n$ . Since  $Z \subseteq J_n$ ,  $Z$  is finite by [Lemma 2.3.1](#), so there exists a bijection  $g : J_m \rightarrow Z$  for some  $m \in \mathbb{N}$ .

By [Theorem 2.2.5](#),  $f|_Z^Y$  is bijective. Then, by [Theorem 2.2.7](#), the composition  $f|_Z^Y \circ g : J_m \rightarrow Y$  is bijective. Thus  $Y$  is finite. □

**Theorem 2.3.3.** *Let  $X$  be a finite ordered set. Then  $X$  has both a least element and a greatest element.*

*Proof.* Since  $X$  is finite, there exists a bijection  $f : J_n \rightarrow X$  for some  $n \in \mathbb{N}$ . The elements of  $X$  can be listed as  $X = \{a_1, a_2, \dots, a_n\}$ , where  $a_k = f(k)$  for each  $k = 1, 2, \dots, n$ .

First, we will show that  $X$  has a least element by mathematical induction on  $n$ . Let  $P(n)$  : Any finite ordered set of cardinality  $n$  has a least element.

- **Base case:** Let  $n = 1$ . Then  $X = \{a_1\}$ , which has a least element  $a_1$ .

- **Inductive step:** Assume that  $P(n)$  is true for some  $n \in \mathbb{N}$ . Let  $X = \{a_1, a_2, \dots, a_{n+1}\}$ . By the inductive hypothesis, the subset  $\{a_1, a_2, \dots, a_n\}$  has a least element, say  $a_k$  for some  $k = 1, 2, \dots, n$ . We have two cases:
  - If  $a_k \leq a_{n+1}$ , then  $a_k$  is the least element of  $X$ .
  - If  $a_{n+1} < a_k$ , then  $a_{n+1}$  is the least element of  $X$ .

Thus  $P(n+1)$  is true whenever  $P(n)$  is true.

By the principle of mathematical induction,  $P(n)$  is true for all  $n \in \mathbb{N}$ .

Next, we will show that  $X$  has a greatest element by mathematical induction on  $n$ . Let  $Q(n)$  : Any finite ordered set of cardinality  $n$  has a greatest element.

- **Base case:** Let  $n = 1$ . Then  $X = \{a_1\}$ , which has a greatest element  $a_1$ .
- **Inductive step:** Assume that  $Q(n)$  is true for some  $n \in \mathbb{N}$ . Let  $X = \{a_1, a_2, \dots, a_{n+1}\}$ . By the inductive hypothesis, the subset  $\{a_1, a_2, \dots, a_n\}$  has a greatest element, say  $a_k$  for some  $k = 1, 2, \dots, n$ . We have two cases:
  - If  $a_k \geq a_{n+1}$ , then  $a_k$  is the greatest element of  $X$ .
  - If  $a_{n+1} > a_k$ , then  $a_{n+1}$  is the greatest element of  $X$ .

Thus  $Q(n+1)$  is true whenever  $Q(n)$  is true.

By the principle of mathematical induction,  $Q(n)$  is true for all  $n \in \mathbb{N}$ . □

**Lemma 2.3.4.** Let  $J_m, J_n$  be given for some  $m, n \in \mathbb{N}$ . Then we have the following properties:

- (A) If there exists an injection  $f : J_m \rightarrow J_n$ , then  $m \leq n$ .
- (B) If there exists a surjection  $f : J_m \rightarrow J_n$ , then  $m \geq n$ .
- (C) If there exists a bijection  $f : J_m \rightarrow J_n$ , then  $m = n$ .

This lemma guarantees the uniqueness of the cardinality of a finite set. If  $J_m \sim X$  and  $J_n \sim X$ , then  $J_m \sim J_n$ , which implies that  $m = n$  by (C).

*Proof.* We will prove (A) by mathematical induction on  $m$ , (B) by mathematical induction on  $n$ , and (C) follows from (A) and (B).

- (A) Let  $n \in \mathbb{N}$  be fixed. Let  $P(m)$  : If there exists an injective function  $f_m : J_m \rightarrow J_n$ , then  $m \leq n$ .
  - **Base case:** Let  $m = 1$ . Then  $m \leq n$  holds.

- **Inductive step:** Assume that  $P(m)$  is true for some  $m \in \mathbb{N}$ . If  $n = 1$ , there does not exist an injective function  $f_{m+1} : J_{m+1} \rightarrow J_n$  since  $f_{m+1}(1) \neq f_{m+1}(m+1)$ , where  $P(m+1)$  is true.

Otherwise, suppose that such a function  $f_{m+1}$  exists. We have two cases:

- $f_{m+1}(m+1) = n$ . Let  $f_m \equiv f_{m+1}|_{J_m}^{J_{n-1}}$ . Then  $f_m : J_m \rightarrow J_{n-1}$  is injective by [Theorem 2.2.5](#). By the inductive hypothesis, we have  $m \leq n-1$ , which implies that  $m+1 \leq n$ .
- $f_{m+1}(m+1) = x \neq n$ . Let  $g_n : J_n \rightarrow J_n$  be defined by

$$g_n(y) = \begin{cases} x, & \text{if } y = n \\ n, & \text{if } y = x \\ y, & \text{otherwise} \end{cases}.$$

Then  $g_n$  is bijective. Let  $f_m \equiv (g_n \circ f_{m+1})|_{J_m}^{J_{n-1}}$ . Then  $f_m : J_m \rightarrow J_{n-1}$  is injective by [Theorems 2.2.5](#) and [2.2.7](#). By the inductive hypothesis, we have  $m \leq n-1$ , which implies that  $m+1 \leq n$ .

Thus  $P(m+1)$  is true whenever  $P(m)$  is true.

By the principle of mathematical induction,  $P(m)$  is true for all  $m \in \mathbb{N}$ .

- (B) Let  $m \in \mathbb{N}$  be fixed. Let  $Q(n)$  : If there exists a surjective function  $f_n : J_m \rightarrow J_n$ , then  $m \geq n$ .

- **Base case:** Let  $n = 1$ . Then  $m \geq n$  holds.
- **Inductive step:** Assume that  $Q(n)$  is true for some  $n \in \mathbb{N}$ . If  $m = 1$ , there does not exist a surjective function  $f_{n+1} : J_m \rightarrow J_{n+1}$  since  $f_{n+1}(1)$  is unique, where  $Q(n+1)$  is true.

Otherwise, suppose that such a function  $f_{n+1}$  exists. We have two cases:

- $f_{n+1}(m) = n+1$ . Let  $K_{n+1} := (f_{n+1})^{-1}(\{n+1\}) \subseteq J_m$ . Let  $f_n : J_m \rightarrow J_n$  be defined by

$$f_n(y) = \begin{cases} f_{n+1}(y), & \text{if } y \in J_m \setminus K_{n+1} \\ n, & \text{if } y \in K_{n+1} \end{cases}.$$

Then  $f_n$  is surjective. By the inductive hypothesis, we have  $m-1 \geq n$ , which implies that  $m \geq n+1$ .

- $f_{n+1}(m) = x \neq n+1$ . Let  $g_{n+1} : J_{n+1} \rightarrow J_{n+1}$  be defined by

$$g_{n+1}(y) = \begin{cases} x, & \text{if } y = n+1 \\ n+1, & \text{if } y = x \\ y, & \text{otherwise} \end{cases}.$$

Then  $g_{n+1}$  is bijective. Let  $K_{n+1} := (g_{n+1})^{-1}(\{n+1\}) \subseteq J_m$ . Let  $f_n : J_m \rightarrow J_n$  be defined by

$$f_n(y) = \begin{cases} (g_{n+1} \circ f_{n+1})(y), & \text{if } y \in J_m \setminus K_{n+1} \\ n, & \text{if } y \in K_{n+1} \end{cases}.$$

Then  $f_n$  is surjective by [Theorem 2.2.7](#). By the inductive hypothesis, we have  $m-1 \geq n$ , which implies that  $m \geq n+1$ .

Thus  $Q(n+1)$  is true whenever  $Q(n)$  is true.

By the principle of mathematical induction,  $Q(n)$  is true for all  $n \in \mathbb{N}$ .

□

**Lemma 2.3.5.** *Let  $J_m, J_n$  be given for some  $m, n \in \mathbb{N}$ . Then we have the following properties:*

- (A) *If  $m \leq n$ , then there exists an injection  $f : J_m \rightarrow J_n$ .*
- (B) *If  $m \geq n$ , then there exists a surjection  $f : J_m \rightarrow J_n$ .*
- (C) *If  $m = n$ , then there exists a bijection  $f : J_m \rightarrow J_n$ .*

*This lemma is the converse of [Lemma 2.3.4](#).*

*Proof.* We will prove the lemma by explicitly constructing the functions.

- (A) If  $m \leq n$ , define  $f : J_m \rightarrow J_n$  by  $f(i) = i$  for all  $i \in J_m$ . Then  $f$  is an injection.
- (B) If  $m \geq n$ , define  $f : J_m \rightarrow J_n$  by  $f(i) = i$  for all  $i \in J_n \subseteq J_m$  and  $f(i) = n$  for all  $i \in J_m \setminus J_n$ . Then  $f$  is a surjection.
- (C) If  $m = n$ , define  $f : J_m \rightarrow J_n$  by  $f(i) = i$  for all  $i \in J_m$ . Then  $f$  is a bijection.

□

**Theorem 2.3.6.** *Let  $X, Y$  be finite sets. Then we have the following properties:*

- (A) *If there exists an injection  $f : X \rightarrow Y$ , then  $|X| \leq |Y|$ .*
- (B) *If there exists a surjection  $f : X \rightarrow Y$ , then  $|X| \geq |Y|$ .*
- (C) *If there exists a bijection  $f : X \rightarrow Y$ , then  $|X| = |Y|$ .*

Moreover, the converses of (A), (B), and (C) are also true.

The contraposition of (A) is called the **pigeonhole principle**, which states that if  $|X| > |Y|$ , then there exists no injection  $f : X \rightarrow Y$ .

*Proof.* Let  $|X| = m$  and  $|Y| = n$  for some  $m, n \in \mathbb{N}$ . Then there exist bijections  $f_X : J_m \rightarrow X$  and  $f_Y : J_n \rightarrow Y$ . By Theorems 2.2.3 and 2.2.7, the composition

$$F \equiv (f_Y)^{-1} \circ f \circ f_X : J_m \rightarrow J_n$$

is injective, surjective, or bijective if  $f$  is injective, surjective, or bijective, respectively. Then, by Lemma 2.3.4, we have the desired results:

- (A) If  $f$  is injective, then  $F$  is injective, which implies that  $m \leq n$ .
- (B) If  $f$  is surjective, then  $F$  is surjective, which implies that  $m \geq n$ .
- (C) If  $f$  is bijective, then  $F$  is bijective, which implies that  $m = n$ .

Also, by Theorems 2.2.3 and 2.2.7, the composition

$$f \equiv f_Y \circ G \circ (f_X)^{-1} : X \rightarrow Y$$

is injective, surjective, or bijective if  $G : J_m \rightarrow J_n$  is injective, surjective, or bijective, respectively. Then, by Lemma 2.3.5, we have the desired converses:

- (A) If  $m \leq n$ , then there exists an injective function  $G : J_m \rightarrow J_n$ , which implies that there exists an injective function  $f : X \rightarrow Y$ .
- (B) If  $m \geq n$ , then there exists a surjective function  $G : J_m \rightarrow J_n$ , which implies that there exists a surjective function  $f : X \rightarrow Y$ .
- (C) If  $m = n$ , then there exists a bijective function  $G : J_m \rightarrow J_n$ , which implies that there exists a bijective function  $f : X \rightarrow Y$ .

□

**Lemma 2.3.7.** Let  $X, Y$  be finite sets with  $X \cap Y = \emptyset$ . Then the union  $X \cup Y$  is finite.

*Proof.* Let  $m = |X|$  and  $n = |Y|$ . There exist bijections  $f : J_m \rightarrow X$  and  $g : J_n \rightarrow Y$ . Let  $h : J_{m+n} \rightarrow X \cup Y$  be defined by

$$h(i) = \begin{cases} f(i), & \text{if } i \leq m \\ g(i - m), & \text{if } m < i \leq m + n \end{cases}.$$

Then  $h$  is bijective, which implies that  $X \cup Y$  is finite.

□

**Theorem 2.3.8.** *Let  $X, Y$  be finite sets. Then the union  $X \cup Y$  and intersection  $X \cap Y$  are finite.*

*Proof.* Let  $A, B, C$  be defined by

- $A = X \setminus Y \subseteq X$ .
- $B = Y \setminus X \subseteq Y$ .
- $C = X \cap Y \subseteq X$ .

Then  $A, B, C$  are finite by [Theorem 2.3.2](#) and are disjoint each other. We have the followings:

- $X = (X \setminus Y) \cup (X \cap Y) = A \cup C$ .
- $Y = (Y \setminus X) \cup (X \cap Y) = B \cup C$ .
- $X \cup Y = ((X \setminus Y) \cup (Y \setminus X)) \cup (X \cap Y) = (A \cup B) \cup C$ .

By [Lemma 2.3.7](#),  $X, Y, X \cup Y$  are finite. □

**Theorem 2.3.9.** *Let  $X_1, X_2, \dots, X_n$  be finite sets for some  $n \in \mathbb{N}$ . Then the union  $\bigcup_{k=1}^n X_k$  and intersection  $\bigcap_{k=1}^n X_k$  are finite.*

*Proof.* Since  $\bigcap_{k=1}^n X_k \subseteq X_1$ ,  $\bigcap_{k=1}^n X_k$  is finite by [Theorem 2.3.2](#) for any  $n \in \mathbb{N}$ . Let

$P(n) : \bigcup_{k=1}^n X_k$  is finite for some  $n \in \mathbb{N}$ , given that  $X_1, X_2, \dots, X_n$  are finite sets.

- **Base case:** Let  $X_1$  be a finite set. Then,  $\bigcup_{k=1}^1 X_k = X_1$  is finite. Thus,  $P(1)$  is true.

- **Inductive step:** Assume that  $P(n)$  is true for some  $n \in \mathbb{N}$ .

Let  $X_1, X_2, \dots, X_{n+1}$  be finite sets. Then we have

$$\bigcup_{k=1}^{n+1} X_k = \left( \bigcup_{k=1}^n X_k \right) \cup X_{n+1}.$$

By the inductive hypothesis,  $\bigcup_{k=1}^n X_k$  is finite. Then, by [Theorem 2.3.8](#),  $\bigcup_{k=1}^{n+1} X_k$  is finite. Thus  $P(n+1)$  is true whenever  $P(n)$  is true.

By the principle of mathematical induction,  $P(n)$  is true for all  $n \in \mathbb{N}$ .  $\square$

**Lemma 2.3.10.** *Let  $X, Y$  be finite sets. Then the cartesian product  $X \times Y$  is finite.*

*Proof.* Let  $m = |X|$  and  $n = |Y|$ . Then there exist bijections  $f : J_m \rightarrow X$  and  $g : J_n \rightarrow Y$ . Let  $h : J_{mn} \rightarrow X \times Y$  be defined by

$$h(i) = \left( f\left(\left[\frac{i}{n}\right]\right), g\left(i - n\left(\left[\frac{i}{n}\right] - 1\right)\right) \right) \text{ for all } i \in J_{mn}.$$

Then  $h$  is bijective, which implies that  $X \times Y$  is finite.  $\square$

**Theorem 2.3.11.** *Let  $X_1, X_2, \dots, X_n$  be finite sets for some  $n \in \mathbb{N}$ . Then the cartesian product  $\prod_{k=1}^n X_k$  is finite.*

*Proof.* Let  $P(n) : \prod_{k=1}^n X_k$  is finite for some  $n \in \mathbb{N}$ , given that  $X_1, X_2, \dots, X_n$  are finite sets.

- **Base case:** Let  $X_1$  be a finite set. Then,  $\prod_{k=1}^1 X_k = X_1$  is finite. Thus,  $P(1)$  is true.

- **Inductive step:** Assume that  $P(n)$  is true for some  $n \in \mathbb{N}$ .

Let  $X_1, X_2, \dots, X_{n+1}$  be finite sets. Then we have

$$\prod_{k=1}^{n+1} X_k = \left( \prod_{k=1}^n X_k \right) \times X_{n+1}.$$

By the inductive hypothesis,  $\prod_{k=1}^n X_k$  is finite. Then, by [Lemma 2.3.10](#),  $\prod_{k=1}^{n+1} X_k$  is finite. Thus  $P(n+1)$  is true whenever  $P(n)$  is true.

By the principle of mathematical induction,  $P(n)$  is true for all  $n \in \mathbb{N}$ .  $\square$

**Theorem 2.3.12.** *Let  $X$  be a nonempty set. Then there exists no surjection  $f : X \rightarrow \mathcal{P}(X)$ , where*

$$\mathcal{P}(X) = \{E \mid E \subseteq X\}$$

*is the **power set** of  $X$ . In particular, if  $X$  is finite, then  $|\mathcal{P}(X)| > |X|$ . This theorem is called **Cantor's theorem**.*

*Proof.* Let  $S = \{x \in X \mid x \notin f(x)\}$ . Then  $S \subseteq X$ , which implies that  $S \in \mathcal{P}(X)$ . Suppose that there exists a surjection  $f : X \rightarrow \mathcal{P}(X)$ . Then there exists  $x \in X$  such that  $f(x) = S$ .

If  $x \in S$ , then  $x \notin f(x) = S$ , which is a contradiction. If  $x \notin S$ , then  $x \in f(x) = S$ , which is also a contradiction. Therefore, there exists no surjection  $f : X \rightarrow \mathcal{P}(X)$ .

If  $X$  is finite, by [Theorem 2.3.6](#), there exists no surjection  $f : X \rightarrow \mathcal{P}(X)$  if and only if  $|X| < |\mathcal{P}(X)|$ .  $\square$

## 2.4 Countable and Uncountable Sets

A set  $X$  is said to be **countable** if  $\mathbb{N} \sim X$ . A set  $X$  is said to be **uncountable** if it is neither finite nor countable. A set  $X$  is said to be **at most countable** if  $X$  is either finite or countable.

**Theorem 2.4.1.** *Let  $X$  be an infinite set and  $Y$  be a finite set. Then  $X \setminus Y$  is infinite.*

*Proof.* Suppose  $X \setminus Y$  is finite. Since  $Y$  is finite, by [Theorem 2.3.8](#),  $(X \setminus Y) \cup Y$  is finite. Then, by [Theorem 2.3.2](#),  $X \subseteq (X \setminus Y) \cup Y$  is finite, which contradicts the assumption. Thus,  $X \setminus Y$  is infinite.  $\square$

**Theorem 2.4.2.** *Let  $X$  be an infinite set. If  $X \sim Y$ , then  $Y$  is infinite.*

*Proof.* Suppose  $Y$  is finite. Then there exists a bijection  $f : J_n \rightarrow Y$  for some  $n \in \mathbb{N}$ . Since  $X \sim Y$ , there exists a bijection  $g : X \rightarrow Y$ .

Then the composition  $g^{-1} \circ f : J_n \rightarrow X$  is a bijection, which implies that  $X$  is finite. This contradicts the assumption that  $X$  is infinite. Thus,  $Y$  is infinite.  $\square$

**Lemma 2.4.3.**  *$\mathbb{N}$  is infinite.*

*Proof.* Suppose  $\mathbb{N}$  is finite. Then there exists a bijection  $f : J_n \rightarrow \mathbb{N}$  for some  $n \in \mathbb{N}$ . Then, by [Theorem 2.3.2](#),  $f(J_n)$  is finite. By [Theorem 2.3.3](#),  $f(J_n)$  has a greatest element, say  $m$ .

Then we have  $m+1 \in \mathbb{N}$  and  $m < m+1 \notin f(J_n)$ , which contradicts the assumption that  $f$  is surjective. Thus,  $\mathbb{N}$  is infinite.  $\square$

**Theorem 2.4.4.** *Every countable set is infinite.*

*Proof.* Let  $X$  be a countable set. Then there exists a bijection  $f : \mathbb{N} \rightarrow X$ . By Theorem 2.4.2 and lemma 2.4.3,  $X$  is infinite.  $\square$

*Remark.* Let  $X$  be a countable set. Then there exists a bijection  $f : \mathbb{N} \rightarrow X$ . The elements of  $X$  can be listed as  $X = \{x_1, x_2, x_3, \dots\}$ , where  $x_n = f(n)$  for all  $n \in \mathbb{N}$ . In other words, the elements of  $X$  can be arranged in a sequence.

**Theorem 2.4.5.**  $\mathbb{Z}$  is countable.

*Proof.* Let  $f : \mathbb{N} \rightarrow \mathbb{Z}$  be defined by

$$f(n) = \begin{cases} -\frac{n}{2}, & \text{if } n \text{ is even} \\ \frac{n-1}{2}, & \text{if } n \text{ is odd} \end{cases}.$$

We will show that  $f$  is bijective, which implies that  $\mathbb{Z}$  is countable.

- **Injectivity:** Suppose  $f(n_1) = f(n_2)$  for some  $n_1, n_2 \in \mathbb{N}$ . Since  $f(n) \geq 0$  for odd  $n$  and  $f(n) < 0$  for even  $n$ , we have  $n_1$  and  $n_2$  are either both even or both odd.
  - If both  $n_1$  and  $n_2$  are even, then  $-\frac{n_1}{2} = -\frac{n_2}{2}$ , which implies  $n_1 = n_2$ .
  - If both  $n_1$  and  $n_2$  are odd, then  $\frac{n_1-1}{2} = \frac{n_2-1}{2}$ , which implies  $n_1 = n_2$ .

Thus,  $f$  is injective.

- **Surjectivity:** Let  $k \in \mathbb{Z}$ . We need to find an  $n \in \mathbb{N}$  such that  $f(n) = k$ .
  - If  $k \geq 0$ , let  $n = 2k + 1$ . Then  $f(n) = \frac{n-1}{2} = \frac{2k}{2} = k$ .
  - If  $k < 0$ , let  $n = -2k$ . Then  $f(n) = -\frac{n}{2} = -\frac{-2k}{2} = k$ .

Thus,  $f$  is surjective.

Since  $f$  is both injective and surjective, we have  $f$  is bijective.  $\square$

**Lemma 2.4.6.** Every infinite subset of  $\mathbb{N}$  is countable.

*Proof.* Let  $X \subseteq \mathbb{N}$  be an infinite subset. Let  $X_1 := X$  and define a function  $f : \mathbb{N} \rightarrow X$  as follows:

- $f(n) = \min X_n$

- $X_{n+1} := X_n \setminus \{f(n)\}$

for each  $n \in \mathbb{N}$ . By [Theorem 2.4.1](#),  $X_n$  is infinite for each  $n \in \mathbb{N}$ , so is nonempty. Then, by [Theorem 1.1.1](#),  $X_n$  has a least element, so  $f$  is well-defined. We will show that  $f$  is bijective, which implies that  $X$  is countable.

- **Injectivity:** Suppose  $n_1 \neq n_2$  for some  $n_1, n_2 \in \mathbb{N}$ . Without loss of generality, assume that  $n_1 < n_2$ . Then  $f(n_1) \in X_{n_1}$  and  $f(n_2) \in X_{n_2}$ . Since  $X_{n_2} \subseteq X_{n_1} \setminus \{f(n_1)\}$ , we have  $f(n_2) \neq f(n_1)$ . Thus,  $f$  is injective.
- **Surjectivity:** Let  $x \in X$ . Let  $S := \{n \in \mathbb{N} \mid n < x\}$ . If  $S = \emptyset$ , then  $f(1) = x$ . Otherwise, by [Lemma 2.3.1](#),  $S \subseteq \{1, 2, \dots, x-1\}$  is finite. Let  $k = |S|$ ,  $S = \{s_1, s_2, \dots, s_k\}$  with  $s_1 < s_2 < \dots < s_k$ . Then we have

- $\min X_1 = \min S \implies X_2 = X \setminus \{s_1\}$ .
- $\min X_2 = \min S \setminus \{s_1\} \implies X_3 = X \setminus \{s_1, s_2\}$ .
- $\vdots$
- $\min X_k = \min S \setminus \{s_1, s_2, \dots, s_{k-1}\} \implies X_{k+1} = X \setminus S$ .

Thus,  $f(k+1) = \min X_{k+1} = x$ . Therefore,  $f$  is surjective.

□

**Theorem 2.4.7.** *Every infinite subset of a countable set is countable.*

*Proof.* Let  $X$  be a countable set and let  $Y \subseteq X$  be an infinite subset. Since  $X$  is countable, there exists a bijection  $f : \mathbb{N} \rightarrow X$ . By [Theorem 2.2.5](#),  $f|_{f^{-1}(Y)} : f^{-1}(Y) \rightarrow Y$  is bijective. Then, by [Theorem 2.4.2](#),  $f^{-1}(Y) \subseteq \mathbb{N}$  is infinite.

By [Lemma 2.4.6](#),  $f^{-1}(Y)$  is countable. Then there exists a bijection  $g : \mathbb{N} \rightarrow f^{-1}(Y)$ . The composition  $f|_{f^{-1}(Y)} \circ g : \mathbb{N} \rightarrow Y$  is bijective, which implies that  $Y$  is countable. □

**Theorem 2.4.8.** *Let  $X$  be infinite. If there exists a surjection  $f : \mathbb{N} \rightarrow X$ , then  $X$  is countable.*

*Proof.* For each  $k \in X$ , let  $S_k := f^{-1}(\{k\})$ . Since  $f$  is surjective,  $S_k \neq \emptyset$  and by [Theorem 1.1.1](#),  $S_k$  has a least element for all  $k \in X$ . Let  $g : X \rightarrow \mathbb{N}$  be defined by

$$g(k) = \min S_k \text{ for all } k \in X.$$

Suppose  $g(k_1) = g(k_2)$  for some  $k_1, k_2 \in X \implies \min S_{k_1} = \min S_{k_2}$ . Since  $S_i \cap S_j = \emptyset$  for all  $i, j \in X$  with  $i \neq j$ , we have  $S_{k_1} = S_{k_2} \implies k_1 = k_2$ . Thus,

$g$  is injective.

Since  $g$  is injective, by [Theorem 2.2.5](#),  $g|^{g(X)} : X \rightarrow g(X)$  is bijective. Since  $X$  is infinite,  $g(X) \subseteq \mathbb{N}$  is infinite by [Theorem 2.4.2](#). Then, by [Lemma 2.4.6](#),  $g(X)$  is countable, which implies that  $X$  is countable.  $\square$

**Theorem 2.4.9.** *Let  $\{X_n\}_{n \in \mathbb{N}}$  be a sequence of countable sets. Then  $\bigcup_{n=1}^{\infty} X_n$  is countable, i.e., any countable union of countable sets is countable.*

*Proof.* Since each  $X_n$  is countable, there exists a bijection  $f_n : \mathbb{N} \rightarrow X_n$  for each  $n \in \mathbb{N}$ . For each  $n \in \mathbb{N}$ , we define the followings:

- $t_n = \min T_n$ , where  $T_n := \left\{ k \in \mathbb{N} \mid n \leq \frac{k(k+1)}{2} \right\}$ .
- $s_n = n - \frac{t_n(t_n - 1)}{2} = n - \frac{t_n(t_n + 1)}{2} + t_n \in \mathbb{N}$ .

Since  $2n \in T_n$ ,  $T_n \neq \emptyset$ . Then, by [Theorem 1.1.1](#),  $T_n$  has a least element. Thus,  $t_n$  is well-defined. Since  $\frac{t_n(t_n - 1)}{2} < n \leq \frac{t_n(t_n + 1)}{2}$ , we have

$$t_n - s_n + 1 = \frac{t_n(t_n + 1)}{2} - n + 1 \geq 1$$

so  $t_n - s_n + 1 \in \mathbb{N}$ . Let  $g : \mathbb{N} \rightarrow \bigcup_{n=1}^{\infty} X_n$  be defined by

$$g(k) = f_{s_k}(t_k - s_k + 1) \text{ for all } k \in \mathbb{N}.$$

Let  $x \in \bigcup_{n=1}^{\infty} X_n$ . Then there exists an  $m \in \mathbb{N}$  such that  $x \in X_m$ . Since  $f_m$  is surjective, there exists a  $l \in \mathbb{N}$  such that  $f_m(l) = x$ . Let

$$k = \frac{(l + m - 1)(l + m - 2)}{2} + m \in \mathbb{N}.$$

We will show that  $l + m - 2 \notin T_k$  and  $l + m - 1 \in T_k$ , which implies that

- $t_k = \min T_k = l + m - 1$ .
- $s_k = k - \frac{t_k(t_k - 1)}{2} = m$ .
- $g(k) = f_{s_k}(t_k - s_k + 1) = f_m(l + m - 1 - m + 1) = f_m(l) = x$ .

Consider the following inequalities when  $z = l + m - 2$  and  $z = l + m - 1$ :

- Let  $z = l + m - 2$ . Then we have

$$\begin{aligned} k - \frac{z(z+1)}{2} &= k - \frac{(l+m-2)(l+m-1)}{2} \\ &= \frac{(l+m-1)(l+m-2)}{2} + m - \frac{(l+m-2)(l+m-1)}{2} \\ &= m > 0. \end{aligned}$$

- Let  $z = l + m - 1$ . Then we have

$$\begin{aligned} k - \frac{z(z+1)}{2} &= k - \frac{(l+m-1)(l+m)}{2} \\ &= \frac{(l+m-1)(l+m-2)}{2} + m - \frac{(l+m-1)(l+m)}{2} \\ &= m - \frac{(l+m-1)(l+m) - (l+m-1)(l+m-2)}{2} \\ &= m - (l+m-1) = 1-l \leq 0. \end{aligned}$$

Thus  $g$  is surjective, which implies that  $\bigcup_{n=1}^{\infty} X_n$  is countable by [Theorem 2.4.8](#).  $\square$

**Corollary 2.4.10.** *Let  $X_1, X_2, \dots, X_n$  be countable sets for some  $n \in \mathbb{N}$ . Then  $\bigcup_{i=1}^n X_i$  is countable, i.e., any finite union of countable sets is countable.*

*Proof.* Let  $X_{n+1}, X_{n+2}, \dots$  be countable sets. Then, by [Theorem 2.4.9](#),  $\bigcup_{n=1}^{\infty} X_n$  is countable. Suppose that  $\bigcup_{i=1}^n X_i$  is finite. Then  $X_1 \subseteq \bigcup_{i=1}^n X_i$  is finite by [Theorem 2.3.2](#), which contradicts [Theorem 2.4.4](#). Thus,  $\bigcup_{i=1}^n X_i$  is infinite. Since  $\bigcup_{i=1}^n X_i \subseteq \bigcup_{n=1}^{\infty} X_n$ , by [Theorem 2.4.7](#),  $\bigcup_{i=1}^n X_i$  is countable.  $\square$

**Theorem 2.4.11.** *Suppose that  $X$  is at most countable, and, for each  $x \in X$ ,  $Y_x$  is at most countable. Then the union*

$$Y = \bigcup_{x \in X} Y_x$$

*is at most countable, i.e., any at most countable union of at most countable sets is at most countable.*

*Proof.* We have 4 cases to consider:

- (A) Finite union of finite sets is finite by [Theorem 2.3.9](#).
- (B) Finite union of countable sets is countable by [Corollary 2.4.10](#).
- (C) Countable union of finite sets is countable.
- (D) Countable union of countable sets is countable by [Theorem 2.4.9](#).

Let us prove (C). Let  $X$  be a countable set and  $Y_x$  be finite for each  $x \in X$ . Then, there exists a bijection  $f : \mathbb{N} \rightarrow X$ , and for each  $x \in X$ , there exists a bijection  $g_x : J_{n_x} \rightarrow Y_x$  for some  $n_x \in \mathbb{N}$ . Let  $h : \mathbb{N} \rightarrow Y$  be defined by

- $h(j) = g_{f(1)}(j)$  for  $j = 1, 2, \dots, n_{f(1)}$ .
- $h\left(\sum_{i=1}^{k-1} n_{f(i)} + m\right) = g_{f(k)}(m)$  for each  $k \in \mathbb{N}$  with  $k \geq 2$ , and  $m = 1, 2, \dots, n_{f(k)}$ .

Then,  $h$  is a surjection from  $\mathbb{N}$  to  $Y$ . By [Theorem 2.4.8](#),  $Y$  is countable.  $\square$

**Lemma 2.4.12.**  $\mathbb{N} \times \mathbb{N}$  is countable.

*Proof.* We will explicitly construct a bijection from  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ . Let  $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$  be defined by

$$f(m, n) = \frac{(m+n-1)(m+n-2)}{2} + m$$

for each  $(m, n) \in \mathbb{N} \times \mathbb{N}$ . We will show that  $f$  is bijective.

- **Injectivity:** Suppose  $(m_1, n_1) \neq (m_2, n_2)$ . If  $m_1 + n_1 = m_2 + n_2$ , then  $m_1 \neq m_2$ , so  $f(m_1, n_1) \neq f(m_2, n_2)$ . Otherwise,  $m_1 + n_1 \neq m_2 + n_2$ . Without loss of generality, assume  $m_1 + n_1 < m_2 + n_2$ . Then, since  $m_1 + n_1 + 1 \leq m_2 + n_2$ , we have

$$\begin{aligned} f(m_1, n_1) &= \frac{(m_1 + n_1 - 1)(m_1 + n_1 - 2)}{2} + m_1 \\ &\leq \frac{(m_1 + n_1 - 1)(m_1 + n_1 - 2)}{2} + (m_1 + n_1 - 1) \\ &= \frac{(m_1 + n_1)(m_1 + n_1 - 1)}{2} \\ &\leq \frac{(m_2 + n_2 - 1)(m_2 + n_2 - 2)}{2} \\ &< \frac{(m_2 + n_2 - 1)(m_2 + n_2 - 2)}{2} + m_2 = f(m_2, n_2), \end{aligned}$$

Thus,  $f(m_1, n_1) \neq f(m_2, n_2)$ , so  $f$  is injective.

- **Surjectivity:** Let  $k \in \mathbb{N}$ . Let  $t_k, s_k$  be defined as in the proof of [Theorem 2.4.9](#). Then we have  $f(s_k, t_k - s_k + 1) = k$ . Thus,  $f$  is surjective.

□

**Theorem 2.4.13.** *Let  $X$  and  $Y$  be countable sets. Then  $X \times Y$  is countable.*

*Proof.* There exist bijections  $f : \mathbb{N} \rightarrow X$  and  $g : \mathbb{N} \rightarrow Y$ . Let  $h : \mathbb{N} \times \mathbb{N} \rightarrow X \times Y$  be defined by

$$h(m, n) = (f(m), g(n)) \text{ for all } (m, n) \in \mathbb{N} \times \mathbb{N}.$$

Then,  $h$  is a bijection from  $\mathbb{N} \times \mathbb{N}$  to  $X \times Y$ . By [Lemma 2.4.12](#),  $\mathbb{N} \times \mathbb{N}$  is countable, so  $X \times Y$  is countable. □

**Theorem 2.4.14.** *Let  $X_1, X_2, \dots, X_n$  be countable sets for some  $n \in \mathbb{N}$ . Then  $\prod_{i=1}^n X_i$  is countable, i.e., any finite product of countable sets is countable.*

*Proof.* We will prove the theorem by mathematical induction on  $n$ . Let  $P(n) : \prod_{i=1}^n X_i$  is countable for some  $n \in \mathbb{N}$ , given that  $X_1, X_2, \dots, X_n$  are countable sets.

- **Base case:** Let  $X_1$  be a countable set. Then  $\prod_{i=1}^1 X_i = X_1$  is countable. Thus,  $P(1)$  is true.

- **Inductive step:** Assume that  $P(n)$  is true for some  $n \in \mathbb{N}$ .

Let  $X_1, X_2, \dots, X_{n+1}$  be countable sets. Then we have

$$\prod_{i=1}^{n+1} X_i = \left( \prod_{i=1}^n X_i \right) \times X_{n+1}$$

By the inductive hypothesis,  $\prod_{i=1}^n X_i$  is countable. Then, by [Theorem 2.4.13](#),

$\prod_{i=1}^{n+1} X_i$  is countable. Thus,  $P(n+1)$  is true when  $P(n)$  is true.

By the principle of mathematical induction,  $P(n)$  is true for all  $n \in \mathbb{N}$ . □

**Theorem 2.4.15.**  *$\mathbb{Q}$  is countable.*

*Proof.* Any  $q \in \mathbb{Q}$  can be expressed as  $q = \frac{m}{n}$  for some  $m \in \mathbb{Z}$  and  $n \in \mathbb{N}$ . Thus, we have

$$f : \mathbb{Z} \times \mathbb{N} \rightarrow \mathbb{Q}, \quad f(m, n) = \frac{m}{n} \text{ for all } (m, n) \in \mathbb{Z} \times \mathbb{N}.$$

is a surjection. By [Theorems 2.4.5](#) and [2.4.13](#),  $\mathbb{Z} \times \mathbb{N}$  is countable. Then, there exists a bijection  $g : \mathbb{N} \rightarrow \mathbb{Z} \times \mathbb{N}$ . Let  $h \equiv f \circ g : \mathbb{N} \rightarrow \mathbb{Q}$ . By [Theorem 2.2.7](#),  $h$  is a surjection.

Since  $\mathbb{N} \subseteq \mathbb{Q}$ , by [Theorem 2.3.2](#) and [lemma 2.4.3](#),  $\mathbb{Q}$  is infinite. Then, by [Theorem 2.4.8](#),  $\mathbb{Q}$  is countable.  $\square$

**Theorem 2.4.16.** Let  $A := \{0, 1\}$ . Then  $X = \prod_{n=1}^{\infty} A$  is uncountable.

*Proof.* For each  $n \in \mathbb{N}$ , let  $e_n \in X$  be the sequence whose  $n$ -th term is 1 and all other terms are 0. Let  $f : \mathbb{N} \rightarrow X$  be defined by  $f(n) = e_n$  for all  $n \in \mathbb{N}$ . Since  $f|_{f(\mathbb{N})}$  is bijective,  $f(\mathbb{N}) \subseteq X$  is countable, so is infinite by [Theorem 2.4.4](#).

Then, by [Theorem 2.3.2](#),  $X$  is infinite. Suppose that  $X$  is countable. Then, there exists a bijection  $g : \mathbb{N} \rightarrow X$ . Let  $X = \{g(1), g(2), g(3), \dots\}$ , where

- $g(1) = (a_{11}, a_{12}, a_{13}, \dots)$ .
- $g(2) = (a_{21}, a_{22}, a_{23}, \dots)$ .
- $\vdots$

Let  $x = (b_1, b_2, b_3, \dots) \in X$  be defined by  $b_n = 1 - a_{nn}$  for all  $n \in \mathbb{N}$ . Then,  $x \neq g(n)$  for all  $n \in \mathbb{N}$ , which contradicts the assumption that  $g$  is surjective. Thus,  $X$  is uncountable.  $\square$

**Theorem 2.4.17.**  $\mathbb{R}$  is uncountable.

*Proof.* Since  $\mathbb{N} \subseteq \mathbb{R}$ ,  $\mathbb{R}$  is infinite, as like [Q is infinite](#). Let  $f : \mathbb{N} \rightarrow [0, 1)$  be defined by  $f(n) = \frac{1}{n+1}$  for all  $n \in \mathbb{N}$ .  $[0, 1) \subseteq \mathbb{R}$  is infinite as like [X is infinite](#). We will prove that  $[0, 1) \subseteq \mathbb{R}$  is uncountable. Then, by [Theorem 2.4.7](#),  $\mathbb{R}$  is uncountable.

Let  $g : [0, 1) \rightarrow \prod_{n=1}^{\infty} \{0, 1\}$  be defined by  $g(x) = \{a_n^x\}_{n=1}^{\infty}$  for all  $x \in [0, 1)$ , where

$\{a_n^x\}_{n=1}^{\infty}$  is the greedy base 2 expansion of  $x$ . Let  $Y := g([0, 1)) \subseteq \prod_{n=1}^{\infty} \{0, 1\}$ .

Then, by [Theorems 1.3.9 and 2.2.5](#),  $g|_Y$  is bijective.

Suppose that  $[0, 1)$  is countable. Then there exists a bijection  $h : \mathbb{N} \rightarrow [0, 1)$ . By [Theorem 2.2.7](#),  $g|_Y \circ h : \mathbb{N} \rightarrow Y$  is bijective, so  $Y \subseteq \prod_{n=1}^{\infty} \{0, 1\}$  is countable. Let

$Z := \prod_{n=1}^{\infty} \{0, 1\} \setminus Y$ . By [Theorems 1.3.10 and 1.3.11](#), we have

$$Z = \left\{ \{a_n\}_{n=1}^{\infty} \in \prod_{n=1}^{\infty} \{0, 1\} \mid a_n = 1 \text{ for all } n \geq N \text{ for some } N \in \mathbb{N} \right\}.$$

Let  $Z_N$  be defined by

$$Z_N = \left\{ \{a_n\}_{n=1}^{\infty} \in \prod_{n=1}^{\infty} \{0, 1\} \mid a_n = 1 \text{ for all } n \geq N \right\} \text{ for each } N \in \mathbb{N}.$$

Let  $F_N : \prod_{n=1}^{N-1} \{0, 1\} \rightarrow Z_N$  be defined by

$$F_N \left( \{a_n\}_{n=1}^{N-1} \right) = \{a_n\}_{n=1}^{N-1} \cup \{1\}_{n=N}^{\infty} \text{ for all } \{a_n\}_{n=1}^{N-1} \in \prod_{n=1}^{N-1} \{0, 1\}.$$

Then  $F_N$  is bijective, so  $Z_N$  is finite by [Theorem 2.3.11](#). By [Theorem 2.4.11](#),  $Z = \bigcup_{N=1}^{\infty} Z_N$  is countable. Then, by [Corollary 2.4.10](#),  $\prod_{n=1}^{\infty} \{0, 1\} = Y \cup Z$  is countable, which contradicts [Theorem 2.4.16](#). Thus,  $[0, 1)$  is uncountable.  $\square$

## 2.5 Metric Spaces

## 2.6 Compact Sets

## 2.7 Perfect Sets

## 2.8 Connected Sets

## **Chapter 3**

# **Numerical Sequences and Series**

### **3.1 Convergent Sequences**

### **3.2 Subsequences**

### **3.3 Cauchy Sequences**

### **3.4 Upper and Lower Limits**

### **3.5 Some Special Sequences**

### **3.6 Series**

### **3.7 Series of Nonnegative Terms**

### **3.8 The Number $e$**

### **3.9 The Root and Ratio Tests**

### **3.10 Power Series**

**3.11 Summation by Parts**

**3.12 Absolute Convergence**

**3.13 Addition and Multiplication of Series**

**3.14 Rearrangements**

## **Chapter 4**

# **Continuity**

### **4.1 Limits of Functions**

### **4.2 Continuous Functions**

### **4.3 Continuity and Compactness**

### **4.4 Continuity and Connectedness**

### **4.5 Discontinuities**

### **4.6 Monotonic Functions**

### **4.7 Infinite Limits and Limits at Infinity**

## **Chapter 5**

# **Differentiation**

### **5.1 The Derivative of a Real Function**

### **5.2 Mean Value Theorems**

### **5.3 The Continuity of Derivatives**

### **5.4 L'Hospital's Rule**

### **5.5 Derivatives of Higher Order**

### **5.6 Taylor's Theorem**

### **5.7 Differentiation of Vector-valued Functions**

## **Chapter 6**

# **The Riemann-Stieltjes Integral**

**6.1 Definition and Existence of the Integral**

**6.2 Properties of the Integral**

**6.3 Integration and Differentiation**

**6.4 Integration of Vector-valued Functions**

**6.5 Rectifiable Curves**

## **Chapter 7**

# **Sequences and Series of Functions**

**7.1 Discussion of Main Problem**

**7.2 Uniform Convergence**

**7.3 Uniform Convergence and Continuity**

**7.4 Uniform Convergence and Integration**

**7.5 Uniform Convergence and Differentiation**

**7.6 Equicontinuous Families of Functions**

**7.7 Equicontinuous Families of Functions**

**7.8 The Stone-Weierstrass Theorem**

## **Chapter 8**

# **Some Special Functions**

### **8.1 Power Series**

### **8.2 The Exponential and Logarithmic Functions**

### **8.3 The Trigonometric Functions**

### **8.4 The Algebraic Completeness of the Complex Field**

### **8.5 Fourier Series**

### **8.6 The Gamma Function**

## **Chapter 9**

# **Functions of Several Variables**

**9.1 Linear Transformations**

**9.2 Differentiation**

**9.3 The Contraction Principle**

**9.4 The Inverse Function Theorem**

**9.5 The Implicit Function Theorem**

**9.6 The Rank Theorem**

**9.7 Determinants**

**9.8 Derivatives of Higher Order**

**9.9 Differentiation of Integrals**

## **Chapter 10**

# **Integration of Differential Forms**

### **10.1 Integration**

### **10.2 Primitive Mappings**

### **10.3 Partitions of Unity**

### **10.4 Change of Variables**

### **10.5 Differential Forms**

### **10.6 Simplexes and Chains**

### **10.7 Stokes' Theorem**

### **10.8 Closed Forms and Exact Forms**

### **10.9 Vector Analysis**

## **Chapter 11**

# **The Lebesgue Theory**

### **11.1 Set Functions**

### **11.2 Construction of the Lebesgue Measure**

### **11.3 Measure Spaces**

### **11.4 Measurable Functions**

### **11.5 Simple Functions**

### **11.6 Integration**

### **11.7 Comparison with the Riemann Integral**

### **11.8 Integration of Complex Functions**

### **11.9 Functions of Class $\mathcal{L}^2$**

# Bibliography

- [1] Walter Rudin, *Principles of Mathematical Analysis*, McGraw-Hill International Editions.
- [2] Kenneth Ross, *Elementary Analysis*, Springer Nature.
- [3] Steven Douglass, *Introduction to Mathematical Analysis*, Pearson.
- [4] Terence Tao, *Analysis I*, Springer Nature.